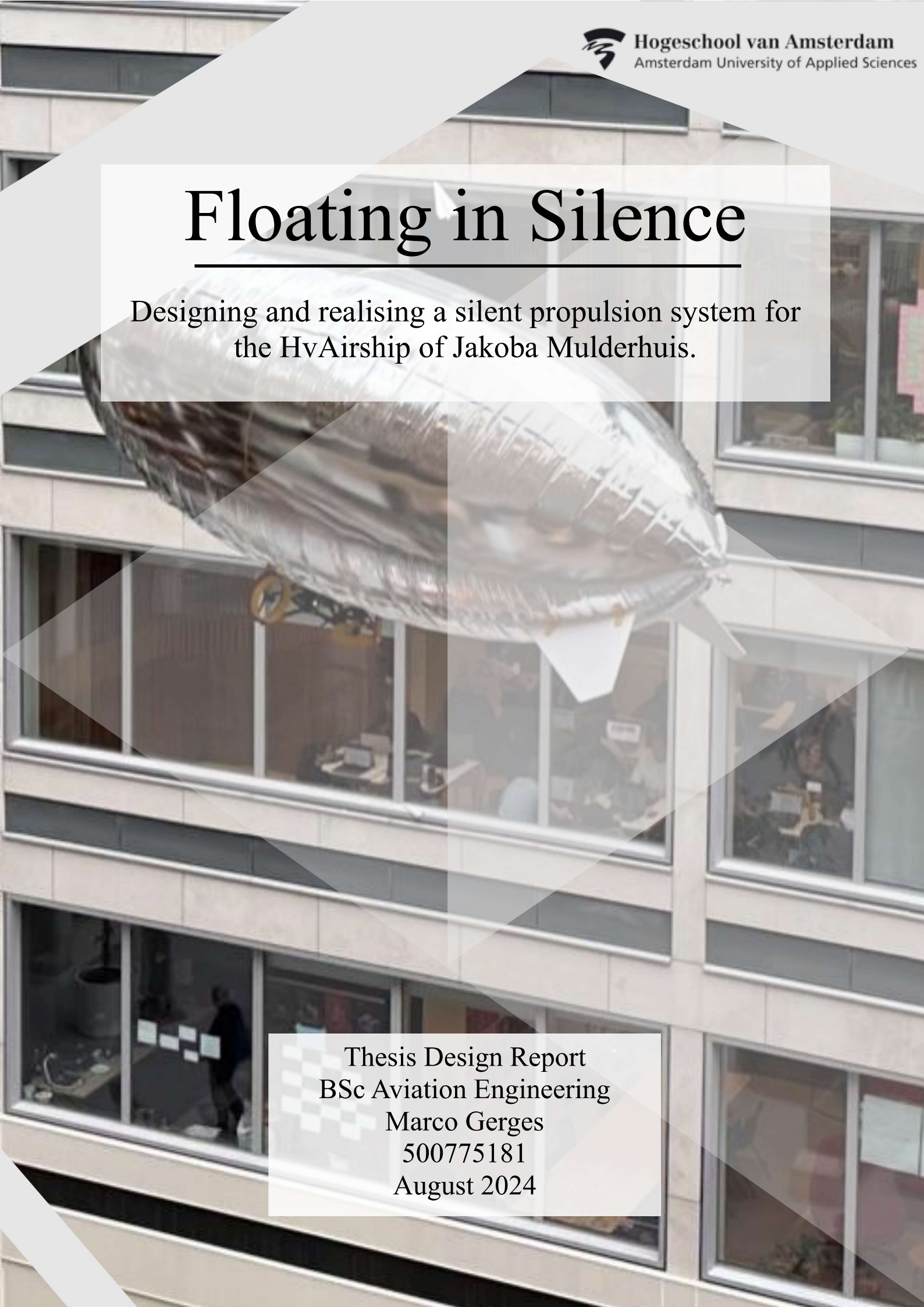
Hogeschool van Amsterdam
Amsterdam University of Applied Sciences

Floating in Silence

Designing and realising a silent propulsion system for the HvAirship of Jakoba Mulderhuis.



Thesis Design Report
BSc Aviation Engineering

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Cleaning in Silence

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Preface

This project has been one of the most challenging projects I have faced. The large amount of disciplines practiced during this project challenged my knowledge and made for a humbling experience.

It is said that science is an iterative process with people building upon the shoulders of giants whom themselves were carried by their own heroes. This experience has truly shown that any project is a collective effort. The amount of people I should be grateful for is humbling. First of all, a sincere display of gratitude has to be shown to the lab lead of the Maintenance Lab Doctor Jurjen Helmus, who not only allowed me to perform my thesis project for the maintenance lab but showed supreme patience with my often-negative outlook on progress during this project.

The awesome team in the maintenance lab were a true delight to work with and taught me a lot simply by performing their projects and allowing my curiosity. In particular Donna Metzlar, who spent hours helping me order anything required for the project, Roald Teunissen and Arthur van Schravendijk, whom both would come with creative solutions to problems encountered during the project, and Daan van Wijck Jurriaanse, who helped with practical portions of the project, are all heroes in my eyes.

Raymond Astudillo van Eijk, the original creator of the HvAirship, is owed gratitude. Not only for his creation which gave me a challenging yet fun project to work on for my thesis but also for his helpful demeaner, his willingness to provide resources and his knowledge.

My lecturer Doctor Maria Papanikou, I have to thank for her frankly herculean effort in reigning in my melancholy. Without her academic knowledge and helpful attitude, I would not have been able to perform the academic portion of this project. A heartfelt thanks is also owed to Doctor Christiaan Schoemaker and lecturer Arthur Bandell who both guided me through the challenging technical and aerodynamic portions of this project. Mister Bandell allowed me to borrow a test bench which was crucial in determining the thrust performance of motors assessed during this project, spent hours discussing issues with me and sent me to professionals who could help me find a solution to a nasty predicament. Doctor Schoemaker spent hours trying to help with complex aerodynamic issues which would have scuttled the project if not for him.

Sincerely,
Marco Gerges

Management Summary

The Amsterdam University of Applied Sciences (AUAS) faculty of technology was moved to the Jakoba Mulderhuis (JMH) from the Leeuwenburg (LWB) building in 2022. The building consists of two sections, the section of interest is the ground floor up till and including the sixth floor. This area, called the atrium, has a wide-open section which stretches all six floors and students have used this to throw paper aircraft. A solution to the problem caused by these paper aircraft was found when the faculty photographer designed and built the HvAirship.

The issue with the HvAirship is that it accumulated a large number of complaints about noise. Therefore, a redesign was required to make it more quiet. When IP-projects were proposed for year 2024-2025 involving the HvAirship, the redesign effort moved up in priority. The objective also shifted from only the design of a less noisy propulsion system on the HvAirship. To a quieter and lighter propulsion system which allows for students to modify the platform for projects.

The achievement of the objective requires a research question which helps guide the report in the right direction. The main research question can be stated in the following form:

'What changes can be made to the propulsion system of the HvAirship drone for it to be quieter while at least maintaining, if not improving its manoeuvrability?'

This project was performed in five phases, orientation, analysis, design, detailing and realisation. In the orientation phase the current system and the principles behind it were explored as well as the context of the project itself and the expectations for the HvAirship. During the analysis phase crucial calculations were performed to determine the design requirements set for the new propulsion system. In this phase it was determined that:

1. A total weight for the propulsion system of less than 500 grams (g) to allow for the minimal 500 g lifting capacity.
    a. Preference for the weight of the propulsion system to be as low as possible.
2. The ability to move the HvAirship forward, backwards, up and down.
3. Total thrust of 150 g to allow for velocities of 3 meters per second (m/s) horizontally and 1.5 m/s vertically/strafing.
4. Low power consumption and excess battery capacity to allow for modifications and longer loiter times.
5. Lower or equal noise levels in db to the lowest mean ambient noise level, 55 db.
6. The system needs to be scalable due to the need for multiple HvAirships for IP projects.
7. The system needs to be safe to operate in JMH.

With these design requirements the propulsion system could be designed. A design based on the Yinyan micro 1104hm 4300 K_V brushless DC motor combined with a GemFan 51-millimetre diameter four bladed propeller to provide the thrust required. The motors will be vectored using servos.

This was followed by the detailing phase where specific subsystems were selected for the propulsion system to be remotely operated. Finally, in the realisation phase the system is to be built and a plan to operate the system is proposed.

Nomenclature

Abbreviation	Elaboration
AUAS	Amsterdam University of Applies Sciences
BLDC	BrushLess Direct Current
BOM	Bill of Materials
CAA	Computational Aeroacoustics
CAD	Computer Assisted Drawings
C_D	drag coefficient
CHC	Center for Hearing and Communication
db	decibels
EMTVP	Envelope-Mounted Thrust Vectoring Pods
ESC	Electronic Speed Controller
FBD	Free Body Diagram
FFT	fast Fourier transform
FMS	Fixed Motor System
FMTVP	Frame-Mounted Thrust Vectoring Pods
g	grams
Hz	Hertz
J	Joules
JMH	Jakoba Mulderhuis
Li-Po	Lithium Polymer
LTA	Lighter Than Air
LTAV	Lighter Than Air Vehicles
LWB	Leeuwenburg
m	metres
m/s	metres per second
mAh	milli Ampère hour
mm	millimetres
N	Newton
PEGASES	Plasma propulsion with Electronegative gases
PR	Powerplant Replacement
RC	Remote Control
RPM	Rotations Per Minute
S	Cells
s	seconds
UAV	Unmanned Aerial Vehicles
V	Volts
VCNS	Virtual Community Noise Simulator
W	Watts

List of Definitions

Term	Definition
Envelope	The part of a lighter-than-air-system which contains the gas used to attain buoyancy.
Gore	A triangular shaped area, which is part of an airship's envelope.
HvAirship	Lighter than air unmanned aerial system. Specifically referring to the system which is to be modified.
K_V	A constant for rotations per minute per Volt.
Measurement set-ups	Constructions with which data of interest can be measured.
Stakeholder Analysis	The identification of and analysis of their power and interest in a project.

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1. Introduction

This report aims to design and field a working propulsion system. This chapter introduces the background and the problem (1.1). Sets an objective (1.2) for the report while grounding the research relevance (1.3). Lastly, scope (1.4) is determined, and deliverables (1.5) are identified.

1.1 Background and Problem

The Amsterdam University of Applied Sciences (AUAS) faculty of technology was moved to the Jakoba Mulderhuis (JMH) from the Leeuwenburg (LWB) building in 2022 (*The Future of Creative Hubs: Jakoba Mulderhuis | Marc Koehler Architects*, 2022). The building consists of two sections, the section of interest is the ground floor up till and including the sixth floor. This area has a wide-open section which stretches all six floors and students have used this to throw paper aircraft. These paper aircraft get stuck in hard-to-reach places, for example window frames, windowsills, and structural beams of the connected building.

Tentative solution and its shortcomings

The faculty has decided to budget a drone which will fly around clearing the debris off of those hard-to-reach locations. This should lower the cost per clean-up relative to hiring cleaning crews. The design selected for this purpose is an airship. The airship, which will be referred to as 'HvAirship' for the remainder of this report, however, has a problem with its propulsion system.

Grounding

The problem with the HvAirship's current propulsion system, is that complaints were submitted by students and faculty members when the drone was in operation. The complaints were related to the noise created when the current propulsion system was engaged. The HvAirship was moved to the maintenance lab at the fourth floor in JMH. With a request for the redesign of the propulsion system in a more silent configuration.

1.2 Objective

The objective of the report is the design and delivery of a less noisy propulsion system on the HvAirship. Additional objectives are the testing and operationalisation of the prototype. To achieve these goals multiple intermediate objectives are required. These objectives are paralleled with design phases retrieved from (Oskam et al., 2017).The results can be seen in table 1.

Intermediate objectives	Oskam et al. phase
Understand the assignment and context	Phase I: Orientation
Define design requirements	Phase II: Analysis
Select design approach and direction	Phase III: Design
Comparison and Testing of Propulsion Systems	Phase IV: Detailing
Building propulsion system and installation	Phase V: Realisation

Table 1: Intermediate objectives with the related design phase.

By anchoring the intermediate objectives with the design phases in table 1, the student hopes to avoid pitfalls later in the design process.

1.3 Research Relevance

The AUAS faculty has requested for the HvAirship to become 'a representative portrait of the spirit of the aviation academy' and faculty of technique as a whole. Therefore, the design process will need to be documented, allowing students to learn and reproduce the process in their future endeavours. By carefully mapping out design choices, capabilities, and limitations of the HvAirship itself, future modification can be accommodated for. Allowing the HvAirship to grow along with the knowledge and skill of the student base. Interestingly, no comparison which compares thrust, noise and weight among the propulsion systems which are to be considered for this project, could be found on the internet. Therefore, this project attempts to rectify this issue by testing and comparing the aforementioned attributes.

1.4 Research Questions and Scoping

The achievement of the objective requires a research question which helps guide the report in the right direction. The main research question can be stated in the following form:

'What changes can be made to the propulsion system of the HvAirship drone for it to be quieter while at least maintaining, if not improving its manoeuvrability?'

1.4.1. Sub-questions

To reach a satisfactory answer to this research question multiple sub-questions need to be defined. By accounting for qualitative and quantitative aspects of the sub-questions, methods of answering said sub-questions can be ascertained. The sub-questions can be found in the in table 2 which are then juxtaposed to the Oskam et al. design phases identified in section 1.2.

Sub-question	Oskam et al. design phase
What stakeholders in the HvAirship project can be identified?	I
What are the design requirements for the propulsion system of the HvAirship?	II
What are relevant propulsion methods for the HvAirship while accounting for design requirements?	III
What is the most applicable propulsion method for the current predicament?	IV
How will the designed propulsion system be integrated onto the HvAirship and its functions?	V

Table 2: Sub-questions assigned with quantitative and qualitative aspects and juxtaposed to the identified design phases.

1.4.3. Research Scope

The focus of this research will be focussed on selecting, designing, proposing, building, and installing a propulsion on the HvAirship drone. With each addition to this summation risk increases. Thus, scope has to be managed vigorously.

Assumptions

To achieve this goal, assumptions have to be made to allow for a manageable scope. These assumptions are:

- No systems outside of the propulsion system on the HvAirship are to be considered.
- Costs related to the propulsion system design process will be fully covered by the maintenance lab, including:
  - Materials
  - Components
  - Power
- The delivered system can be considered a one-off design.
  - Thus, no scale issues have to be considered.
  - The design will require longevity meaning that the system must be maintainable for the coming years.

Inclusions

Testing the noise and thrust capabilities of each propulsion system is deemed to be a critical part of the research. Making sure the system remains controllable is deemed as an integral component of the delivery of a working design.

Exclusions

Creating an autonomous guidance system will be deemed out of scope. Improving its debris clearing capabilities may be considered as out of scope as well. There is no expectation that the propulsion system is the first in a series and thus the resulting system can be treated as a prototype.

Operational scope

This project will have a duration of twenty weeks starting from 1 April 2024 and lasting till 19 August. The project will take place in the maintenance lab and other spaces in the JMH. Due to the project lasting through the summer break planning will be compressed and work tempo will be increased to allow for the project to be completed by 19 July 2024. This negates the possible effects which may be experienced if the JMH is closed down during the summer break. Giving an actual ‘active’ project duration of fifteen weeks. The remaining five weeks will be used to refine the thesis report and prepare for the defence of the thesis.

1.5 Deliverables

This project will have to comply with AUAS Graduation thesis guidelines. Therefore, the deliverables will include a full thesis report as well as a functioning propulsion system design to be installed on the HvAirship.

2. Theoretical Framework

This Chapter aims to review available literature on airships and remote-controlled platforms (2.1). This is followed with an in-depth review of current propulsion technologies (2.2). Finalised with a discussion of methods which may help inform the approaches of the sub-questions (2.3).

2.1 Platform Orientation

The first step towards modifying a system is to know the foundation upon which the system is based and the design of the system itself.

2.1.1. Lighter than Air Vehicles

The HvAirship uses a balloon, envelope, filled with helium gas to float in the air. This method of subverting gravity is based on the Archimedes' principle, also known as buoyancy. The principle states the following:

$$F_b = -\rho g V \tag{1}$$

The principle founds the basis of the idea that a matter which is less dense than the surrounding matter can create a force which carries the less dense matter above the environmental matter. In layman's terms, this makes a system which uses this principle a 'Lighter Than Air' (LTA) vehicle.

LTA vehicles (LTAV) operate in force equilibrium. Contrary to fixed-wing and rotary-wing aircraft, LTAVs do not require lift to be generated actively. In figure 1 a simplified Free Body Diagram (FBD) represents the forces acting on an indoor balloon.

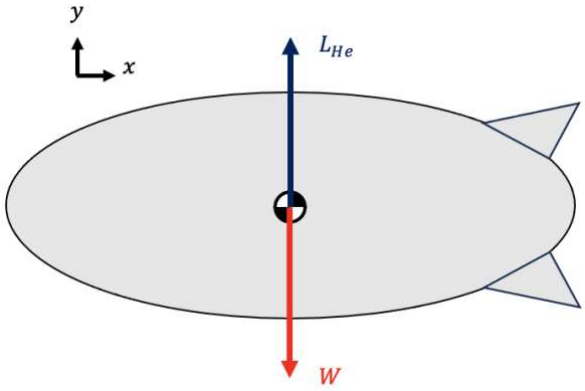


Figure 1: A simplified FBD representing the vertical forces (y-axis) acting on an LTAV.

Objects traveling through fluids are subject to fluid dynamics. Since LTAVs travel through air, LTAVs are subject to the drag equation.

$$D = \frac{1}{2} * c_D * \rho * v^2 * S \tag{2}$$

The drag equation depends on the drag coefficient of the LTAV envelope. This drag coefficient can be estimated based on a figure consulted from a table showing general ranges for C_D (Aziz et al., 2008), figure 2.

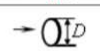
Type of body		Reference area S	Reynolds number Re	Drag coefficient C_D
Cube		$S = D^2$	$Re > 10^4$	1.05
		$S = D^2$	$Re > 10^4$	0.8
Solid hemisphere		$S = \pi D^2 / 4$	$Re > 10^4$	$\rightarrow 0.42$ $\leftarrow 1.17$
Hollow hemisphere		$S = \pi D^2 / 4$	$Re > 10^4$	$\rightarrow 0.38$ $\leftarrow 1.42$
Thin disk		$S = \pi D^2 / 4$	$Re > 10^3$	1.1
Circular disk		$S = \pi D^2 / 4$	$Re \leq 1$ $Re > 10^4$	20.4/Re 1
Sphere		$S = \pi D^2 / 4$	$Re \leq 1$ $1 < Re \leq 2\times10^5$ $Re > 2\times10^5$	24.0/Re 0.45 0.2
Streamlined body		$S = \pi D^2 / 4$	$Re > 10^5$	0.04

Figure 2: Drag coefficients for certain 3d shapes (Aziz et al., 2008).

No literature was found discussing the acoustic properties of indoor LTAVs. This aspect will thus be treated as a blind spot during the analysis phase.

Martinez (2024) created a website with calculations and background information on propellers. This source will deepen the level of understanding of the working principles and mechanics of propellers. The understanding will then be used in the design phase (Oskam et al., 2017)to select the optimal propeller set up if required (Martinez, 2023).
He also created a webpage with calculations and background information on propulsion systems as a whole. This source will help guide the design phase (Oskam et al., 2017) onto the correct design direction while selecting a propulsion method for the HvAirship.

2.1.2. HvAirship

The propulsion system which will need to be modified is attached to and controlling a lighter-than-air envelope designed and manufactured by Windreiter. The specific envelope, in layman's terms the balloon portion of a blimp, is the 4-gore elliptical balloon. This means that the envelope has four gores. Gores are defined as triangular shapes. Windreiter uses the Series 58 Model Gertler 4621 for calculations, shown in appendix A and figure 3. However, Gertler (1950)shows a completely different shape with a more elongated tail. Watt (Watt et al., 2006) correctly complies with the shapes shown in Gertler 1950 which raises concerns for the accuracy of the calculations performed by Windreiter in the accompanying datasheet. Aerodynamic data consulted from the data sheet will thus be considered not fully trustworthy and ignored.

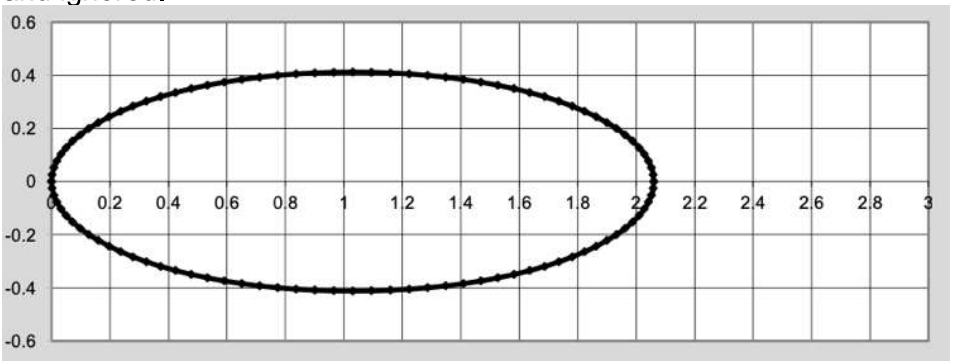


Figure 3: A graph consulted from Windreiter HQ's data sheet which shows a side profile of the envelope.

The shape of the envelope is maintained by the pressure differential between the lifting gas, in this case helium, and the atmosphere (Liao & Pasternak, 2009).Thus, no structural components are present within the envelope itself. Therefore, the HvAirship can be classified

to be a non-rigid airship which is also known as a blimp. The HvAirship is not capable of autonomous flight and thus is remotely controlled. In table 3 a list of all sub-systems contained in the current system is provided.

Sub-system	Model	Price
Frame	Japalura Frame Kit - Choose 3" or 4"	€ 79,00
BrushLess Direct Current (BLDC) motor	Armattan oomph 1407/3850 kv	€ -
Flight Controller	Matek H743-SLIM	€ 119,00
Electronic Speed Controller (ESC)	AIKON AK32PIN 35A 2-6S 4-IN-1 ESC V2	€ 56,50
Radio receiver	TBS Crossfire Nano RX SE	€ 33,95
Camera transmitter	TBS UNIFY PRO 5G8 HV RACE	€ 30,95
Video receiver	TBS Triumph Pro 5.8GHz FPV Antenne - RHCP - U.fl	€ 19,95
Camera	RUNCAM EAGLE 2	€ 39,00
Remote controller	TBS Tango	€ -
Envelope	4-Gore Elliptical Balloon	€ 78,64

Table 3: List of subsystems installed on HvAirship and respective prices.

Mister van Eijk provided a Fusion 360 file with his original design for the HvAirship. In figures 4, 5 and 6, these Computer Assisted Drawings (CAD) can be viewed.



Figure 4: A diagonal profile of the original HvAirship design.



Figure 5: A frontal profile of the original HvAirship design.



Figure 6: A side profile of the original HvAirship design.

It was however, discovered that the envelope was incapable of carrying this payload and as such some parts were cut off or downsized. The modified propulsion payload can be viewed in figure 7.

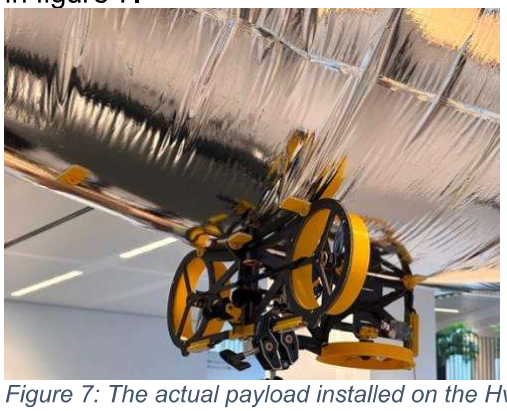


Figure 7: The actual payload installed on the HvAirship envelope.

As can be seen in figure 7. The front section of the payload, including the strafing motor and propeller, was removed. The camera was re-attached to the payload. To the back of the payload the Lithium Polymer (Li-Po) battery was removed along with the entire frame behind the vertical motor and propeller mount. This ensured that the HvAirship remained positively buoyant and thus capable of flight. The HvAirship can be seen while flying in figure 8.



Figure 8: A picture of the HvAirship while in flight.

The initial configuration operates on INAV 7.0.0 which is an open-source Navigation-enabled flight control software. This system allows the payload to be controlled and certain subsystems to be configured. For example, the BLDC motors installed, can be configured to operate with lower power outputs than the maximum allowed. This is performed in the mixer, provided by the INAV application. The mixer can be viewed in its initial configuration in figure 9. Figure 9 is accompanied by table 4. Which describes the most important settings.

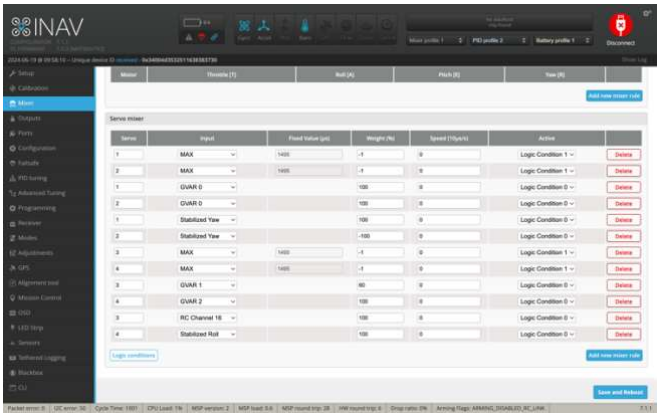


Figure 9: A screenshot of the initial configuration programmed for HvAirship's flight controller.

Input name	Result:
GVAR 0	Forward and backward propulsion
Stabilized Yaw	Yaw steering
GVAR 1	Vertical propulsion
GVAR 2	Strafing propulsion (not installed)

Table 4: Inputs with resulting output manoeuvres.

Since the HvAirship project was performed with great speed. No time was allocated to lower the maximum power output in the horizontal plane. This may be a path to lowering noise levels emitted from the HvAirship.

According to the datasheet provided by Windreiter HQ, appendix A the envelope provided can carry 600.4 grams (g) at sea level. This number will be utilised for initial calculations.

2.2 Noise and weight reduction methods

The prior sub-chapter delves into the system and its founding principles. The system, however, needs to be modified due to its excessive weight and noise. Reduction of these factors can be achieved with multiple approaches.

2.2.1. Noise physics and modelling

Envia, Wilson and Huff (2004) discuss the unresolved issues for Computational Aeroacoustics (CAA) regarding the modelling of fan noise. This paper will be instrumental in understanding the issues which are associated with modelling fan noise. The understanding of these issues will help avoid these issues over the course of the design process.

Brungarta, Olsonb, Klinec and Yoas (2019) show 'that significant reductions in propeller noise can be achieved by increasing propeller diameter while reducing its rotational speed, thereby reducing its blade tip speed, while maintaining a given level of static thrust. Fan affinity laws are also used to show that efficiency improvements accompany the reductions in radiated noise'. The methodology for reducing fan noise will be crucial in the orientation phase, while the noise measurement method will be copied for the design process.

Miller and Sullivan (1985) created a preliminary design tool for propellers combining a fast vortex lattice aerodynamic analysis, fast sub-sonic point source noise analysis and an optimization scheme using a conjugate directions method. This memorandum will be used to help design new propellers if such a step is deemed as required during the design process.

Haddaoui (2019) models noise effects using Virtual Community Noise Simulator (VCNS). This thesis attempts to develop a source noise model for propellers which has the future goal of being integrated in a tool like VCNS or an aircraft design chain. The method of noise measurement is of interest for this design project. The method will be used as a guideline for the test rig which will be set up to measure the noise levels and directions of the tested propulsion systems.

Though these four papers provide ample knowledge on the noise characteristics of propulsion systems. It is surmised that the size of the platform does not allow for fan noise to be a significant factor in noise reduction, especially when taking weight constraints into account. Therefore, Envia et. al (2004) is not considered to be of use for this particular project. Brungarta et. al (1985) seems to be the most promising due to its fundamentally sound approach to propeller noise. The tool created by Miller and Sullivan has not been made available to the AUAS and thus cannot be considered as a viable route. The same logic disqualifies VCNS.

2.2.2. Alternative propulsion methods

Kim (Kim, 2016) conducted an experimental study on the noise characteristics of four-bladed general aviation propellers with uneven blade spacing. The use of uneven blade spacing created additional tones over which the acoustic intensity was distributed. The propulsion method propelled, and its noise characteristics are of considerable interest as an alternative to conventional propeller set-ups. Considering noise properties determined in this thesis helps substantiate decisions during the design process.

Massachusetts Institute of Technology (MIT) (2022) created an info sheet describing the characteristics of the toroidal propeller design. The information displayed in this info sheet will help substantiate design choices made.

Sebastian and Strem (2020), the inventors of the toroidal propeller design, submitted a patent through MIT for a toroidal propeller. This alternative propulsion system is of interest during the design phase. The noise characteristics for this propulsion system at its reduced weight might potential resolve weight issues originating from the lighter than air platform on which the propulsion system will have to be installed.

Kasteloo (2023) wrote an article about the fascinating zipline asymmetric propeller, explaining its innerworkings. This alternative propulsion system is of interest during the design phase. The noise characteristics for this propulsion system at its reduced weight might potential

resolve weight issues originating from the lighter than air platform on which the propulsion system will have to be installed.

Seeley (2019)submitted a patent through the for a conventional ultra-quiet propeller system. The noise characteristics for this propulsion system at its reduced weight might potential resolve weight issues originating from the lighter than air platform on which the propulsion system will have to be installed.

Aanesland, Meige and Charbert (2009) proposed the use of both positive and negative ions for thrust in an electromagnetic space propulsion system. This concept is called PEGASES for Plasma propulsion with Electronegative gases and has been patented by the Ecole Polytechnique in France in 2007. The paper reviews important work that has been used in the process of designing the first PEGASES prototype. The thesis will consider the Ion-ion thrust system outlined and designed in this paper as one potential propulsion method.

A propulsion method which has not yet been mentioned, is the so-called ‘air swimmer’ option. This system depends on the buoyancy of the balloon to imitate fish swimming in the ocean. However, no literature was found discussing this propulsion method.

2.3 Design Process

Beard and McLain (2012) wrote a book explaining the theory and practice behind designing a small, unmanned aircraft. The design principles and fundamentals discussed will be of great interest while creating a viable design for the propulsion system of the HvAirship

Oskam et al. (2017) wrote a book in Dutch explaining the design process in detail. This book will be the bedrock of the design process outlined for the thesis project. Especially the design phases are utilised to direct sub-questions in a direction which achieves optimal results.

3. Methodology

This Chapter considers the design of the project. It consists of a section which considers the phases and tools (3.1) which are to be used. Consequently, a process diagram is created (3.2).

3.1 Phasing and Tools

In Chapter 1 sub-section 1.4.1., the sub-questions were defined. These sub-questions, however, require background questions to be answered. This section discusses the background questions which steer the process in the correct direction. The design project will, as stated in section 1.2., consist of five phases. These are:

I. Orientation
II. Analysis
III. Design
IV. Detailing
V. Realisation

The questions identified for these phases require the usage of several tools and analyses. These tools are listed in this section. Phases are identified by their purpose, and do not reflect chronology. Phases may overlap or be revisited over the course of the design process. This will allow for iterative changes to be made to the design based on feedback.

Phase I

Due to the spotted history of the HvAirship, a need has been identified to identify stakeholders in the HvAirship project. By identifying stakeholders and assessing their interest juxtaposed against their power over the project, importance can be assigned to the design requirements set by each stakeholder. For phase I, a stakeholder analysis will have to be performed to identify who the stakeholders are and their influence over the design project and thesis. No alternative methods for doing so, were identified. The research tools which are to be used are:

- Interviews
    - Identifying stakeholders which may have been missed initially.
- Stakeholder matrix
    - Identifying the influence of stakeholders and determining approach towards them.

Phase II

After identifying the stakeholders. Interviews will be conducted with the most important stakeholders to clarify design requirements. The most important stakeholders are the creator of the HvAirship, maintenance lab representatives and project partners. Since these parties directly provide crucial information about potential design requirements, system functions, design constraints. If a need arises for large numbers of people in a single group to be consulted, surveys are to be utilised to achieve this. These tools are to be utilised specifically with the goal of identifying design requirements. With the data gathered from these tools an assessment will be performed to identify the most important design parameters.

Noise measurements will be taken inside the JMH. These measurements will be taken from multiple locations in the building, on multiple times with and without the airship flying. The first measurement method will be performed with a basic decibel scale sound level meter to measure the ambient and HvAirship decibel levels. A second measurement method will use a sensitive microphone to collect data which can feed a Fast Fourier Transform (FFT).

The decibel noise level meter will also be used to test the noise transmitted from the initial propulsion system in the lowest viable output levels programmed in INAV. Viable meaning that the airship still remains controllable. These will give a clear picture of the state of the current system and its shortcomings regarding noise.

All components of the HvAirship will be carefully weighed to compare to the extrapolated design requirements.

Phase III
Desk research will be utilised to identify the advantages and disadvantages of proposed propulsion systems and configurations. For designs Fusion 360 will be used to create technical drawings. The technical drawings will be made of the proposed propeller configurations, alternative propulsion systems and the testing set-ups. The tools which are to be used are:

- Fusion 360
- Google Sketchup

Phase IV
Data will be collected from the measurement set-ups and this data will be analysed using mathematical and computational tools. It is assumed that Fourier transforms will need to be used for vibrational and noise analysis. If so, the preferred platform for this analysis will be Python or R. It is assumed that Fourier transforms will need to be used for vibrational and noise analysis. If so, the preferred platform for this analysis will be Python or R. MATLAB is not capable of bringing capabilities to the table which are not provided by Python and or R. Additionally, no license for MATLAB is provided for this project. Therefore, due to cost constraints and negligible advantages provided by it, MATLAB will not be considered. An off the shelf option for the control system will be considered. If this is not possible a control program will have to be built manually, but this is considered to be high risk and inefficient.

An alternative programme for modelling noise intensity and propagation is COMSOL. This system is extremely impressive when consulting the company website and could be of great use for the project. The issue with this system is that it is prohibitively expensive, a license can exceed the total budget set for the HvAirship project. The financial constraint thus, like MATLAB , prohibits the use of this software.

Phase V
The final phase will be geared towards physical installation of the system, testing of the control system for the new propulsion system, stakeholder approval and planning for future usage. Due to the sensitivity of the envelope and electronics involved it is presumed an iterative approach will be required. Focussing on the last point, maintenance planning and operational procedures will be proposed. Due to the size of the platform on which the propulsion system will be installed a storage plan will likely be required. Additionally, due to the importance of this phase it is expected that there will be multiple iterations before a final configuration is built.

3.2 Methodology Diagram

To create a clear picture of the project's usage of certain tools a diagram is created. This diagram will show the chronology, relations, and goals for the tools. Showing when a tool is used and why. The diagram is accompanied by a legend which can be found in table 5 and appendix B:

Abbreviation	Elaboration
I1	Interviews round 1
SA	Stakeholder Analysis
I2	Interviews round 2
S	Surveys*
SDR	Setting Design Requirements
SI	Stakeholder Inclusion
D	Design
P	Prototyping
PST	Propulsion System Testing
DD	Detail Design
PM	Platform Modification
FT	Functional Tests

Table 5: Legend for the process diagram.

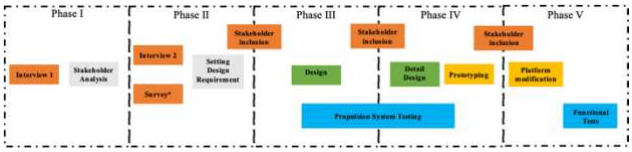


Figure 10: Design process diagram

Note that the performance of a survey will only occur when a need is identified during the stakeholder analysis.

4. Orientation

This chapter considers the first phase of this design project, namely orientation. This subject is tackled with an acknowledgement of technical data provided at the start of the project (4.1), interviews (4.2) and a summary of email correspondence with Facility Services (FS) through intermediaries (4.3).

4.1 Technical Specifications

The first step of the design/modification process is orientation. This means that the platform which is to be modified will need to be fully understood. Due to the lack of documentation on the HvAirship, this section extracts information based on measurements and the scarce documentation found.

Documentation

In appendix A, a spreadsheet page can be found in which the technical weight characteristics can be found. Based on interviews conducted and initial calculations estimates of HvAirship's new design requirements can be set.

4.2 Interviews

First interviews were conducted to allow stakeholders to introduce themselves and divulge their relation to the HvAirship. These interviews' purpose is also to extract information about the stakeholders themselves minding the guidelines set in Murray-Webster & Simon (2006). These guidelines state that the interview will have to steer towards the identification of the stakeholders' interest in the project, power over the project and attitude towards the project.

Interviews will be conducted in a semi-structured format as per the guidelines set in Dipboye (1994). This will ensure that as much information will be extracted while keeping the subject on topic.

The interviews will result in tables per stakeholder which assess interest in, power over, and attitude towards the project. These assessments and the definitions can be viewed in table 6.

Aspect	Assessment	Definition
Interest	High	The stakeholder is very interested in the project due to the project's effects on their work, environment, or financial investment.
	Neutral	The stakeholder is tangentially interested in the project but does not perceive to have an active stake in the project.
	Low	The stakeholder has no interest in the project.
Power	High	The stakeholder has executive influence over the project meaning that the project can be cancelled on their directive.
	Intermediate	The stakeholder can influence the project's direction.
	Low	The stakeholder cannot influence the project, nor can they influence its direction.
Attitude	Positive	The stakeholder has a favourable view of the project.
	Neutral	The stakeholder does not care about the project.
	Negative	The stakeholder shows animosity towards the project.

Table 6: Definitions assigned to assessments for stakeholder analysis considerations.

First stakeholder

The first stakeholder identified is the AUAS photographer for the technique faculty who initially designed and built the HvAirship. His name is Raymond Astudillo van Eijk and will from now on be referred to as mister van Eijk. The main conclusions are listed below:

- Mister Van Eijk will be handing over complete control over the airship to the Maintenance Lab. Making doctor Jurjen Helmus, the lab lead, custodian of the HvAirship.
- HvAirship was built over the course of the 2023-2024 winter recess with an especially purchased envelope, cannibalised quadcopter drone components and a self-designed and -built carbon fibre construction.
- There is no documentation related to the construction process except for the order for the envelope and the control chip.
- A Bill of Materials (BOM) will be created and declared by mister van Eijk with the purpose of financial compensation in exchange for custody of the HvAirship shifting to the Maintenance Lab.
- Based on van Eijk's interpretation of the purpose of the HvAirship maximising loiter time should be the dominant design requirement.

With this information conclusions regarding the aforementioned three main aspects can be assessed in table 6:

Aspect	Assessment
Interest	High
Power	Low
Attitude	Positive

Table 7: Mister van Eijk's stakeholder assessment.

Judging from these values mister van Eijk can be classified as a *friend* (Murray-Webster & Simon, 2006) meaning that he as an insignificant active backer.

Second stakeholder

The second stakeholder is the lab lead of the Maintenance Lab, Doctor Jurjen Helmus. Doctor Helmus has initiated the HvAirship modification project and is the current custodian. Therefore, the information attained from this interview will be crucial for the stakeholder stage and the determination of design requirements. The main conclusions are listed below:

- Doctor Helmus has confirmed that he is the main custodian of the HvAirship.
- The main focus of the Maintenance Lab is to research autonomous inspection opportunities. This is also the point of view from which the HvAirship will be assessed.
- The purpose envisaged for the HvAirship is as follows:
    - HvAirship should become a learning object.
        - Especially for Aviation students.
    - The platform should enable easy modification.
        - The report should enable this point from a documentation perspective.
    - The system should be:
        - Modular
        - Minimal in weight
        - Potentially able to increase lifting capacity.
        - Able to fly silently or with minimal noise disruption.
        - Able to remain in the air for longer periods of time.
            - Especially in comparison to equivalent quadcopter drones.
- Doctor Helmus admitted that he has more projects to manage and thus the HvAirship project will not be focused on from an attention point of view.
- Doctor Helmus does see an opportunity for this project to help strengthen relations with the Aviation faculty of AUAS and thus, has a very positive outlook on the project and the relations with faculty staff it enables.

Based on the data collected from the interview the three main aspects can be assessed in table 7:

Aspect	Assessment
Interest	Neutral
Power	High
Attitude	Positive

Table 8: Doctor Helmus stakeholder assessment.

Due to Doctor Helmus' neutral interest, he can be assessed to be in between of an influential active backer and its passive analogue (Murray-Webster & Simon, 2006). It is judged to be prudent to keep him informed over the course of the project, while making sure that his needs are actively managed and met.

Third stakeholder

Luke Oversluizen is an electrical engineering student at AUAS who is currently following a machine learning minor. The minor includes in its curriculum a requirement for the performance of a project. The project assigned to Luke is to detect paper airplanes. This leads to his connection to the HvAirship project. Due to his position as a student, he does not have the power to influence the project. With this information conclusions regarding the aforementioned three main aspects can be assessed in table 8:

Aspect	Assessment
Interest	Low
Power	Low
Attitude	Positive

Table 9: Luke Oversluizen's stakeholder assessment.

Therefore, Luke can be treated as an insignificant passive backer (Murray-Webster & Simon, 2006).

Remaining parties

As stated in Section 1.1 AUAS' technique faculty is coalesced in the JMH. The building has innkeepers, who are responsible for the building. Upon contact with the innkeepers, it was stated that they have no interest in the HvAirship project due to it being defined by them as a 'slow moving helium balloon' thus being a low risk for damage to the building or personnel. They did refer to the student dean who, according to mister van Eijk, has given express permission for the HvAirship to fly around in the JMH building as long as it is safe. These parties will thus not be considered to be stakeholders. With this information conclusions regarding the aforementioned three main aspects can be assessed for the remaining parties in table 9:

Aspect	Assessment
Interest	Low
Power	High
Attitude	Neutral

Table 10: Remaining stakeholders' assessment.

Based on this assessment the remaining parties will be classified and treated as influential passive backers (Murray-Webster & Simon, 2006).

The conclusion from the stakeholder analysis is that Doctor Jurjen Helmus' assessed design requirements will be seen as the leading design requirements. These requirements, extracted from the interviews are compiled into the following list:

- Maximizing loiter time and minimizing weight are the leading variables.
    - All variables behind these two variables.
- The optical set-up from Luke Oversluizen's report will need to be accounted for.
- The propulsion and control system will have to be as light weight as possible.

Based on these conclusions, Doctor Helmus was contacted, and new design requirements were discussed. The design requirements are set with the following guidelines:

- A maximum of 3 meters per second (m/s) of horizontal and vertical speed.
- No requirements set for turning speed.
- A loiter time of at a minimum 2 hours.
- An excess carrying capacity of 500 grams (g) for future modifications and payloads.
- A maximum of 200 g allocated for optical system designed by Luke Oversluizen and/or other students.
- A full throttle noise level in decibels (db) which is less than the average ambient noise in JMH.

4.3 Faciliatory Considerations

In initial e-mail correspondence FS of JMH and members of the faculty shared their concerns for the HvAirship project. This section summarises these concerns.

An initial worry was the legality of the HvAirship flying inside the building. Citing guidelines created for drone flying by the Dutch government (Waterstaat, 2020) a list of considerations was shared. However, when these guidelines were read, the guidelines turn out to be for outdoor drones or drones flying in public spaces. Since the JMH cannot be considered a public space, these guidelines cannot be considered as definitive. Several other safety and operational concerns were shared and are listed below.

- Helium expansion risk
    - Concern about the potential for helium to expand due to heating, leading to an explosion.
- Falling objects
- Utility and necessity
    - Questions about the purpose and necessity of the zeppelin.
- Operational duration and checks
    - Clarification needed on whether the zeppelin stays airborne for the entire duration of blocks 1 and 2 and the frequency of condition checks (pressure and battery voltage).
- Calamity response
    - How will the HvAirship respond when it loses altitude.

These concerns will be accounted for while setting design requirements. For the long term for the HvAirship project the FS are identified as the most likely future obstacle to continuation of the project and should be treated as such.

5. Analysis

This chapter tackles the second phase of the design project, namely Analysis. This phase starts with an in-depth analysis of physics and aerodynamics (5.1), environment (5.2) and propulsive performance (5.3). Resulting in combination with the prior chapter in the setting of design requirements (5.4).

5.1 Aerodynamics and Physics

The analysis of the platform will be constituted on 6 points of interest. These points can be surmised in the following list:

1. Aerodynamics and Physics
    a. Aerodynamic calculations for the envelope.
    b. Extrapolating thrust requirements.
2. Environmental factors
    a. Acoustics
        i. Ambient noise measurements
3. Propulsive performance
    a. Thrust
    b. Acoustics
        i. Platform noise measurements
4. Setting design requirements

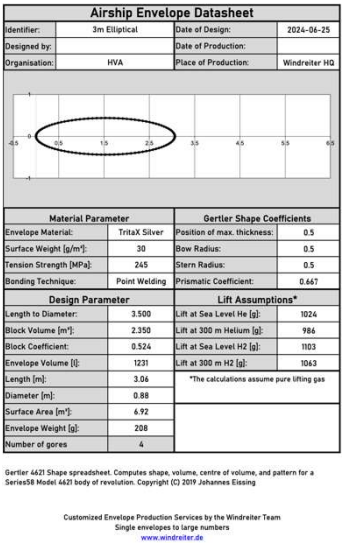
These 5 points determine the sub-sectioning for this section.

5.1.1. Lifting capacity

As can be determined from figure 8, Windreiter has determined a lift at sea level for Helium (He) of 600.40 g. However, Windreiter assumes a purity of 100% for the He supply. AUAS did not use a pure He supply, it sufficed with a 99% pure He supply. Assuming the rest of the gas supply is regular air, a lift at sea level with He is calculated to be 591.39 g. Subtracting the weight of the envelope itself (136.2 g) gives a lifting capacity of 455.2 g. Conversely subtracting 51.6 g for the tailfins attached to the rear of the envelope gives 403.6 g of lifting capacity. Finally, subtracting the weight of the propulsion system and a 4 cell (S) battery (390.6 g) leaves a pitiful 13.0 g of excess weight for sensors or payloads.

When validating the possible excess weight however, it was determined that an excess weight of 43.6 g was available. After consideration, it was determined that on the date of measurement the temperature in JMH was at 23.8 degrees Celsius, it is therefore plausible the temperature being higher than room temperature allowed the lift to increase. During a second validation at 22.8 degrees an excess lifting capacity of 24.68 g was measured.

The maximum measured excess lifting capacity was identified as inadequate. Even with weight saving measures the envelope cannot consistently provide the minimum 500 g of excess lifting capacity required. Conversely, a new envelope was created and provided by Windreiter to AUAS. The new datasheet can be viewed in figure 11 and appendix C.



Airship Envelope Datasheet

Identifier:	3m Elliptical	Date of Design:	2024-06-25
Designed by:		Date of Production:	
Organisation:	HVA	Place of Production:	Windreiter HQ

Material Parameter		Gertler Shape Coefficients	
Envelope Material:	TritaX Silver	Position of max. thickness:	0.5
Surface Weight [g/m²]:	30	Bow Radius:	0.5
Tension Strength [MPa]:	245	Stern Radius:	0.5
Bonding Technique:	Point Welding	Prismatic Coefficient:	0.667

Design Parameter		Lift Assumptions*	
Length to Diameter:	3.500	Lift at Sea Level He [g]:	1024
Block Volume [m³]:	2.350	Lift at 300 m Helium [g]:	986
Block Coefficient:	0.524	Lift at Sea Level H2 [g]:	1103
Envelope Volume [l]:	1231	Lift at 300 m H2 [g]:	1063
Length [m]:	3.06	*The calculations assume pure lifting gas	
Diameter [m]:	0.88		
Surface Area [m²]:	6.92		
Envelope Weight [g]:	208		
Number of gores	4		

Gertler 4621 Shape spreadsheet. Computes shape, volume, centre of volume, and pattern for a Series58 Model 4621 body of revolution. Copyright (C) 2019 Johannes Eissing

Customized Envelope Production Services by the Windreiter Team
Single envelopes to large numbers
www.windreiter.de

Figure 11: Redesigned airship envelope datasheet provided by Windreiter HQ.

5.1.2. Extrapolating thrust requirements

With the redesigned envelope, being deemed acceptable by Doctor Helmus. The next step was performing aerodynamic calculations. The calculations are performed to determine the required thrust to achieve both vertical and horizontal motion.

Aerodynamic method

Based on the drag equation mentioned in chapter 2, the drag coefficient (C_D) will be the deciding variable for determining required thrust. Since the C_D will be higher for larger flatter surfaces (Aziz et al., 2008), it is determined that the drag therefore must be highest during vertical and or sideways i.e. strafing motion. The envelope surface will be similar in both directions of travel due to the elliptical shape of the balloon, therefore vertical and strafing motion will be calculated using the same values.

As a precaution, a correction of 20% to the thrust requirements will be added to make sure any losses due to imperfections in the surface of the airship and the drag introduced with the new propulsion system.

An immediate issue experienced is that no values for drag coefficients can be found for Ellipsoids. During research a particle physics paper by Tran-Cong, Gay and Michaelides (2004) was determined to have an equation which Sadaf Maramizonouz and Sadegh Nadimi (2022) deemed to be “promising”. This equation can be consulted below:

$$C_D = \frac{24}{Re}\left(\frac{d_{surfsph}}{D_{sph}}\right)\left(1 + \frac{0.15}{\sqrt{X_{surf}}}\left(\frac{d_{surfsph}}{d_{sph}}Re\right)^{0.687}\right) + \frac{0.42\left(\frac{d_{surfsph}}{d_{sph}}\right)^2}{\sqrt{X_{surf}\left[1 + 42500\left(\frac{d_{surfsph}}{d_{sph}}Re\right)^{-1.16}\right]}} \quad (3)$$

However, for a velocity of 3 m/s this equation delivered an unrealistic value of 0.72. Based on Aziz et al. (2008) and figure 2 it is estimated that a value between 0.35 and 0.45 is more realistic. Based on the equations related to determining the variables, it is assessed that X_{surf} may not be correct, therefore the decision was made to replace the variable with X. Since both variables denote circularity, the modified equation should remain valid.

$$C_D = \frac{24}{Re}\left(\frac{d_{surfsph}}{D_{sph}}\right)\left(1 + \frac{0.15}{\sqrt{X}}\left(\frac{d_{surfsph}}{d_{sph}}Re\right)^{0.687}\right) + \frac{0.42\left(\frac{d_{surfsph}}{d_{sph}}\right)^2}{\sqrt{X\left[1 + 42500\left(\frac{d_{surfsph}}{d_{sph}}Re\right)^{-1.16}\right]}} \quad (4)$$

The modified equation computes a C_D of 0.37 for a velocity of 3 m/s. This value is within the aforementioned bounds. In appendix D two tables can be viewed which show the resulting C_D for different velocities. From these tables, a horizontal motion with a velocity of 3 m/s requires 150 grams of thrust after correction. An equivalent force in the vertical direction would give a speed of around 1.5 m/s.

Physics method

With uncertainty, introduced by modifying equation (), a validation will be required. This validation can be achieved by using the kinetic energy equation:

$$E_k = {}^1/_2\, mv^2 \quad (5)$$

With this equation kinetic energy can be calculated. Given a velocity of 3 m/s and a mass of 1016 g, which will be the maximum lifting capacity of the new envelope, the kinetic energy is calculated to be 4.6 Joules (J). Using this figure to calculate the power required using the following equation:

$$P = \frac{E_k}{t} \quad (6)$$

By assuming the time used to achieve a velocity of 3 m/s is 3 seconds (s). The equation computes that the power required is 1.5 Watts (W). This number will be used to validate the prior sub-section when power requirements are calculated.

5.2 Environmental factors

As stated in the introduction the HvAirship was grounded due to it being too loud for students and staff in the building. With noise being a leading factor for the clearance of the modified HvAirship to fly in JMH, special considerations are to be enacted. Two methods will be utilised to identify noise in JMH.

1. Noise intensity will be measured in db using a sound level meter.
2. The frequency of the noise will be measured and analysed.

5.2.1. Noise intensity

Using a UNI-T UT353 BT mini sound level meter noise measurements were taken in the JMH. This was performed in two steps. First random sound measurements were taken during random times throughout a day for a week. This is to determine the loudest and quietest days of the week. Following this, these days an hourly measurement will be performed to determine an average and mean intensity during quiet and loud days.

After measuring data randomly throughout the day, it was determined that Tuesdays and Thursdays are the loudest days of the week while Fridays are the quietest. Since the measurements were taken randomly throughout the day and at random positions throughout the building, the data will not be judged to be accurate enough to use for analysis. Therefore, the second step will be performed.

The second step is an hourly measurement from five different positions, from which it is judged the airship will be experienced the most. These locations are as follows:

I. Fourth floor at the E-wing.
   A. This location is deemed to be a prime location for take-offs and landings for the HvAirship.
II. Fourth floor balcony between the E- and D-wings.
   A. This location was chosen due to its proximity to classrooms in the D-wing and the maintenance lab itself.
III. Sixth floor D-wing
   A. This location was chosen to measure noise from the highest accessible level and its presence on the D-wing.
IV. Ground floor middle next to the stairs to -1.
   A. This location was chosen due to its central location and giving a perspective on ground level experience.
V. First floor E-wing in the large open space.
   A. This location was chosen due to its presence directly below the most likely area where the HvAirship will fly.

The data from these measurements can be consulted from appendix E. The measurements were also plotted in figure and figure, for Tuesday and Friday respectively.

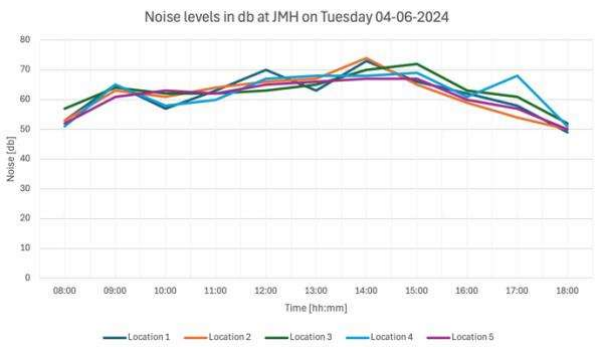


Figure 12: Noise levels in db at JMH on Tuesday 04-06-2024.

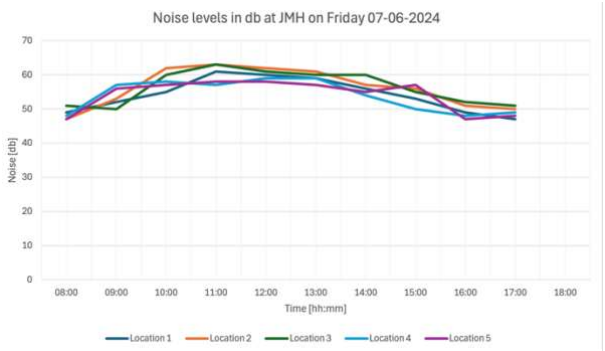


Figure 13: Noise levels in db at JMH on Friday 07-06-2024.

Based on this data tables are compiled with the mean, median, maximum and minimum values.

Mean	Median	Maximum	Minimum
62	63	74	49

Table 11: Main statistical information of sound intensity on Tuesday.

Mean	Median	Maximum	Minimum
55	57	63	47

Table 12: Main statistical information of sound intensity on Friday.

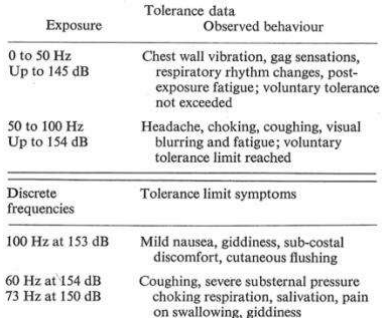
There are however some caveats to consider when reviewing this data. These data points were accumulated during the first week of June. This is in the experience of the student, a slower period in the building due to lower workloads for students in the final block of the second semester and lower numbers of students present in the building due to attrition. It therefore should be considered plausible that this data can be on the quieter side than the first semester.

A more poignant consideration is the fact that these data points seem very high. According to the Center for Hearing and Communication (CHC) in 2024, “Experts agree that prolonged exposure to noise above 70 dB can be harmful to your hearing” (Nadler, 2020). Having a maximum intensity of 74 db seems worrisome. It may be possible that the sound level meter used is overestimating the db level. The manufacturer states that the sound level meter has a basic accuracy of ±1.5 db. This must be considered when reviewing this data by itself. For the context of this research. Since measurements of the sound intensity of the propulsion system will be measured with the same system, it can however be considered together since any likely inaccuracy or overestimation may be equally applied for the propulsion system as the ambient noise.

5.2.2. Noise frequency

Knowing sound intensity however is not the only factor when considering noise complaints. As can be quoted from Berglund, Hassmén and Job (1996), “The primary, and most frequently reported, perceived effect of low-frequency noise is not that of loudness or noisiness, but that of annoyance (Broner, 1978).” Broner (1978) gives the following figure to work with.

These values will be considered when analysing frequency data.



Tolerance data

Exposure	Observed behaviour
0 to 50 Hz Up to 145 dB	Chest wall vibration, gag sensations, respiratory rhythm changes, post-exposure fatigue; voluntary tolerance not exceeded
50 to 100 Hz Up to 154 dB	Headache, choking, coughing, visual blurring and fatigue; voluntary tolerance limit reached
Discrete frequencies	Tolerance limit symptoms
100 Hz at 153 dB	Mild nausea, giddiness, sub-costal discomfort, cutaneous flushing
60 Hz at 154 dB 73 Hz at 150 dB	Coughing, severe substernal pressure choking respiration, salivation, pain on swallowing, giddiness

Figure 14: Tolerance data consulted from Broner (1978).

Frequency data will be accumulated using a RØDE wireless GO II. This microphone has one of the best frequency response curves in the market right now and was available for use during this project. The frequency response curve of the microphone can be viewed in figure 15:

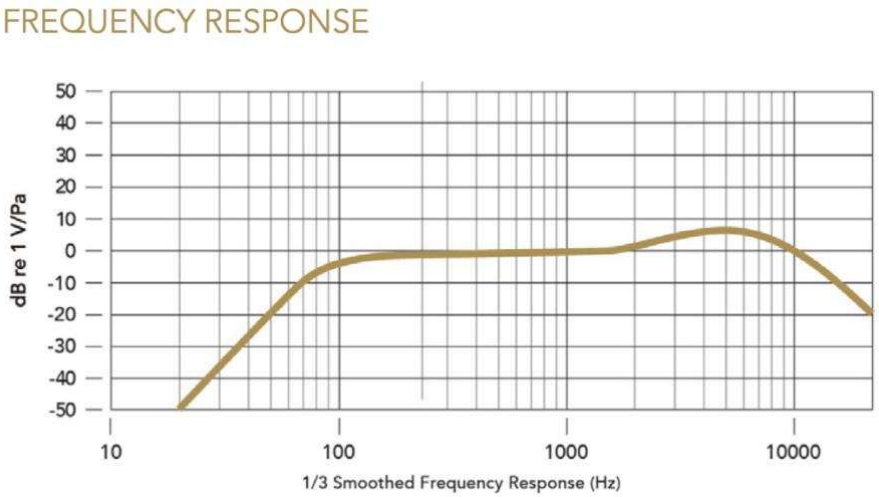


Figure 15: Frequency response curve of the RØDE wireless GO II, consulted from RØDE (2024).

With the microphone providing accurate data, the next step was determining a location in which the microphone could record data without interference for long periods of time. With consultation from FS of JMH, a third-floor balcony which was only accessible for staff was selected. The microphone would make recordings in waveform format. The duration of access to the balcony was set for one week.

Using open-source code (Mulla, 2022). A series of FFT's were created as well as spectrograms. The results can be viewed in the figures below. Figure 16 shows a comparison in FFT, spectrogram and MEL spectrogram forms between ambient noise, the original INAV configuration, a modified INAV configuration with HvAirship in the same position as the original INAV configuration relative to the microphone and the modified INAV configuration from a greater distance. More detailed versions of the graphs are available in appendices F through Q.

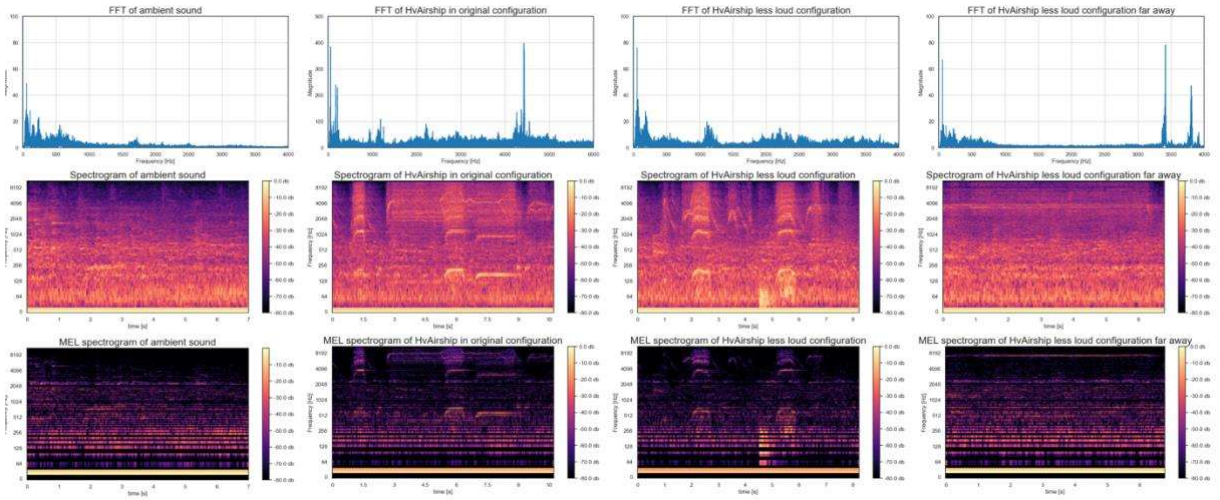


Figure 16: Comparison in FFT, spectrogram and MEL spectrogram forms between ambient noise, the original INAV configuration, a modified INAV configuration with HvAirship in different positions.

As can be seen in the FFTs, ambient sound mostly peaks between a frequency of 0 to 600 Hertz (Hz). The original configuration data shows peaks in the same frequency range notably, these peaks are almost the octuple of the ambient sound magnitude. However significant peaks can be around 4500 Hz, these are likely the noise caused by the BLDC motors. Interestingly, looking at the FFTs of the HvAirship in a less loud configuration, both at close and far range, the pattern of amplified peaks around the same frequencies as the ambient sound with peaks at the higher end of the frequency range.

5.3 Propulsive Performance

Considering the amplified magnitudes emitted by the HvAirship's propulsion system seen in figure 16. The propulsion systems loudness will have to be measured. This was to be performed with the following 3 methods:

1) Noise intensity measurements using the UNI-T UT353 BT mini sound level meter while the HvAirship, with original INAV configuration, is in flight.
2) Noise intensity measurements using the UNI-T UT353 BT mini sound level meter while the HvAirship, with a modified INAV configuration, is in flight.
3) Noise intensity measurements using the UNI-T UT353 BT mini sound level meter with an unattached propulsion system which is kept aloft by wool strings.
4) Noise intensity measurements using the UNI-T UT353 BT mini sound level meter with a detached motor while simultaneously measuring thrust in a testbench.

Technical issues however sabotaged this effort. Due to a leak developing in the envelope and the omission of a reserve envelope, two weeks were spent waiting for a replacement envelope to arrive. Which upon delivery immediately developed a considerable leak. Two more weeks were spent awaiting 5 replacement envelopes which arrived with faulty welds. A further three weeks were spent communicating with Windreiter and negotiating an appropriate form of compensation. This compensation would be the redesigned 3-meter-long elliptical envelopes. By the time the envelopes arrived the summer break was in affect and the propulsion system had been dismantled to maximize usage of the remaining time.

The third method resulted in a BLDC motor being burned through which caused the measurements to be cut short. However, the incomplete data can be viewed in appendix R. A picture of the set-up can be viewed in figure 17. It must be noted that the test was performed at 10% and 100% thrust. When considering prior calculations, a thrust of 26% is more applicable. A possible conclusion from this data may be that the propeller guards, the yellow housing surrounding the propeller, significantly increases the intensity of the sound.



Figure 17: A picture of the noise intensity measurement set-up.

This leaves data accumulated using the fourth method. As compensation for the lack of long-range data a sound intensity measurement was taken from both 0.15 meters and 3 meters of distance from the propeller. This should show a plausible noise intensity drop-off over increasing distances. In figure 18 a picture of the setup can be viewed. The set-up makes use of cloth (socks) and bubble wrap to minimize resonance between the table and microphone set-up. The measurement was performed 3 times to determine a plausible range to take inaccuracies in measurements into account.

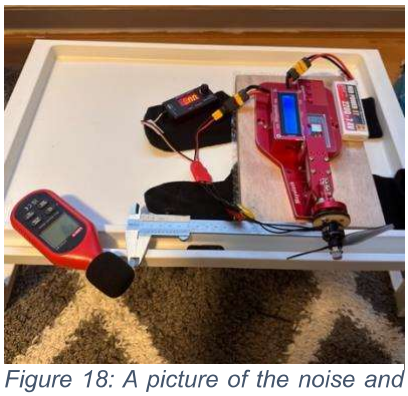


Figure 18: A picture of the noise and thrust measurement set-up.

The results of these measurements can be viewed in appendix S. For the calculated 0.7 Newton (N) of force at 0.15 meters sound intensity ranged between 77.6 and 71.5 db. Close to 100 times the mean sound measured on quiet days at JMH. At three meters sound intensity ranged between 61 and 60.1 db, 1 db less than the mean on busy days at JMH.

5.4 Setting Design Requirements

With the calculations mentioned in the prior section, design requirements for the propulsion system will need to adhere to the following list of requirements. This list of requirements is sub divided in categories to help rate the efficacy of design concepts in section 6.4.

- Thrust
  - Configuration can provide the minimum required 150 grams of thrust.
- Lightweight
  - Will likely be lighter than 500 grams total.
- Omnidirectional
  - Can move in the vertical and/or longitudinal directions from a stationary position.
  - Can rotate on the horizontal plane in place.
- Multipurpose
  - Platform allows for modifications to be installed on the frame.
- Silent
  - Platform minimizes number of noise sources.
  - A powerplant of the system does not exceed a sound intensity of 55 db when exerting total thrust required.
- Power conserving
  - Platform configuration saves energy to ensure loiter time maximization.
- Maintainable
  - Systems can be reached for maintenance and modification without requiring the dismantling of the frame.
- Safe
  - System does not pose a risk to its surroundings, with regard to damage and or injury.
  - No risk of fire is present.
  - No threat of falling objects.
- Producible
  - System can be (re-) created with minimal issues.
  - Components can be replaced with of the shelf replacements.

6. Design

The fourth phase, design, starts with the consideration of all collected propulsion methods (6.1). Followed by an analysis of motor thrust and sound intensity (6.2) and motor selection (6.3). Continued with the proposal of multiple concepts (6.4) and concluded with the selection of the most applicable design (6.5).

6.1 Propulsion System Consideration

Now that design requirements have been determined, which can be found in section 5.4. Propulsion systems can be considered on their merits. The propulsion systems can be categorized in X categories, a list of the categorized propulsion systems can be viewed bellow.

- Unconventionally powered design
  - PEGASES
  - Air swimmer
  - Gas turbine
- Conventionally powered design
  - BLDC motor set-up
    - Using a conventional propeller with larger diameter than motor is designed for.
    - Using a conventional propeller with diameter motor is designed for.
    - Using a cycloidal propeller
    - Using a toroidal propeller
    - Using a X-form propeller
    - Using an asymmetric propeller
  - BDC motor set-up
    - Using a conventional propeller with larger diameter than motor is designed for.
    - Using a conventional propeller with diameter motor is designed for.
    - Using a cycloidal propeller
    - Using a toroidal propeller
    - Using a X-form propeller
    - Using an asymmetric propeller

6.1.1. Unconventionally Powered Systems

Consisting of PEGASES and Air swimmer, these options offer some tempting advantages towards some design requirements but disqualifying properties.

PEGASES

Though ION-thrust offers a very silent solution (Aanesland et al., 2009). Aanesland et. al (2009) express concern for electromagnetic interference caused by PEGASES. Electromagnetic issues contributed to the Hindenburg disaster of 1937 (DiLisi, 2017). Even though, the buoyancy of HvAirship is based on helium and not hydrogen. It is judged that, due to the system flying in a confined space surrounded by electronics, PEGASES is not a safe direction to take this project.

Air swimmer

By far the cheapest, most promising and safest option. Reviewing footage of air swimmers in action reveals issues.

A. Air swimmers need a motor which move the ‘fins.’ These fins are unidirectional which means a fin will need to be placed at every end of every axis.
   a. Considering the three axes and all axes having to directions this means that a total of 6 motors will be required. This will obviously incur a weight penalty.

B. Due to the asymmetric shape of the envelope. The fins will need to have different sizes to compensate for the higher drag encountered in certain directions. Further raising weight.
C. Footage shows air swimmers to have slow velocities. It is doubtful these systems can achieve the required velocities.

Gas turbine
The notion of a floating fuel tank attached to a combustion engine seems to be the least safe option and therefore is not to be considered seriously. Especially when considering the email exchange

6.1.2. Conventionally Powered Systems
Now that unconventional systems have been disqualified conventionally powered propulsion systems remain. Multiple propulsion systems are not widely accessible for public use and must be conversely disqualified. These methods include toroidal, cycloidal, asymmetric (zipline), and Seeley's propellers. While it is possible to design surrogates or imitations, the issue encountered will be the fact that the efficiency of the imitations will be lower than the original. The only method of production available is 3d printing, comparing this method to injection moulding, the resulting propeller design will be heavier, less refined and thus, probably louder due to turbulence caused by the rough and uneven surfaces.

This leaves conventional propellers, of all possible diameters, or x-form asymmetric propellers as the remaining propulsion viable methods.

6.2 Alternative Motors
When considering motors three directions are possible. These three directions are:
1. Larger BLDC motor with lower Rotations Per Minute (RPM) but bigger propellers.
2. Smaller BLDC motor with lower RPM and similar or smaller diameter propellers.
3. BDC motor

In the Remote Control (RC) community and with DC motor manufacturers a constant for rotations per minute per Volt, or K_V, is in use. By finding this value a number of rotations per minute can be calculated. Since the HvAirship's Armattan 00mph 1407 BLDC motors are rated with a K_V of 3850, this means that for a 2S battery voltage of 7.4 Volts (V) the number of RPM is 28490.

Considering the aforementioned 0.7 N required for each of the two motors to achieve the 150 g of thrust, is achieved at 26% of the possible thrust and assuming a linear relation between this percentage and RPM, an RPM of 7407.4 is calculated. Applying the inverse process as the prior sub-sub-section a K_V of ± 1000 is calculated as a possible optimal value.

6.2.1. Larger BLDC Motor
When looking on the internet for BLDC motors with ±1000 K_V one cheap and widely accessible option was found, namely the A2212/13T. This motor is without a doubt the quintessential 1000K_V BLDC motor found in numerous YouTube videos as an example of BLDC motors in general. This motor was purchased for testing and weighing. When the motor was weighed a flaw in the potential motor was discovered. The motor weighed 52.7 g. meaning that a similar set-up as the original propulsion system would weigh 158.1 g. Considering the original motor weighed 15.7 g. This weight, without accounting for propellers, speed controllers, flight controllers and other systems which need to be installed is unacceptable. Therefore, a smaller alternative had to be found. No BLDC motor of equivalent K_V was found.

6.2.2. Similar BLDC Motor with Slightly Lower RPM

Since a 1000 K_V motor was not qualified due to weight constraints a smaller BLDC motor had to be found, this led to the purchase of the ECO Micro Series 1407 motor. This motor has a K_V of 2800, 1050 less than the original motors. Meaning that these motors will likely require a lower voltage/ power setting to operate at the calculated RPM. Since both motors are 1407 type motors, it is assumed that power consumption and voltage calculations are similar and thus approximately, a voltage of 2.6 V or a power setting of 36% is calculated. This motor weighs 15.3 g, which is a minor improvement over the original motors. Some technical information (*★EMAX ECO Micro Series 1407 2~4S 2800KV 3300KV 4100KV Brushless Motor For FPV Racing RC Drone*, n.d.) can be viewed in figure 19 and 20.

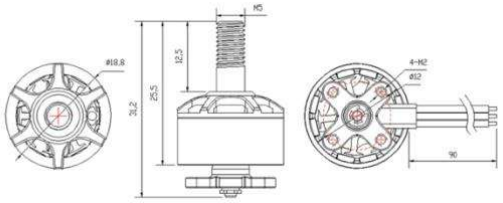


Figure 19: Dimensions of the ECO Micro Series 1407 BLDC motor.

	KV2800	KV3300	KV4100
Idle Current(10V)	0.7A	0.9A	1.1A
Internal Resistance	180mΩ	148mΩ	102mΩ
Peak Current(10S)	13.6A	17.7A	21.6A
Max.Power(10S)	220W	280W	340W
Propeller	3"-4"		3"-3.5"
No.of cells	2-4S		
Lead	26#AWG 90mm		

Figure 20: Technical specifications of ECO Micro series 1407 BLDC motor.

Considering this motor will still be overpowered for the intended thrust an alternative motor which is much smaller has to be found. The motor which has been decided on is the Yinyan Micro 1104 Hm. These motors are rated at 4300 K_V but are miniscule in comparison to all motors considered before. A single motor weighs 5 g, less than a third of the original motors. When looking at an image comparing the four motors sizes a true sense of the size difference can be achieved, as can be seen in figure 21.

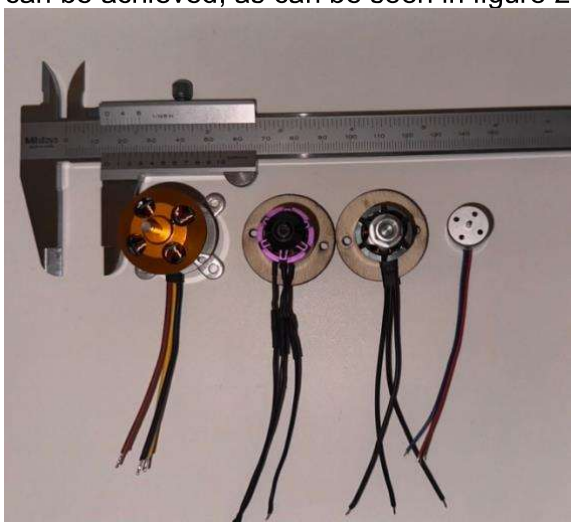


Figure 21: Image with all four BLDC motors together for scale.

6.2.3. BDC Motors

An alternative to BLDC is the less sophisticated BDC motor. These motors do not receive a signal which they translate into a power output or a rotational direction, like their BLDC counterparts, but directly translate the voltage into rotation. This means that the circuits controlling this type of motor need to be more sophisticated and thus will require more weight. However, a small quadcopter drone was purchased to allow for the BDC motors to prove their efficacy.

6.3 Motor and Propeller Selection

To test the motors for thrust and sound intensity, the same method is used to test the new BLDC motors and drone compared to the original system. As can be seen in appendix S. The Yinyan Micro 1104 Hm came out on top. Not only providing adequate thrust with a 51 millimetres (mm) diameter propeller but also meeting the 55 db requirement at 3 m distance. However, it should be noted that ECO Micro series with custom X-form propellers came closest to the thrust and noise intensity requirements. Showing the clear advantages of the X-form principle over conventional four bladed propellers.

However, due to the weight savings and lower sound intensity, the Yinyan Micro 1104 Hm motor is to be selected as the optimal option for this project. The Yinyan Micro 1104 with the GemFan D51 four bladed propellers provides a 0.7 N thrust with 54% thrust, giving adequate margin to account for mistakes of miscalculations regarding thrust requirements.

6.4 Alternative Propulsion System Concepts

Three design concepts are to be considered for the propulsion system. These designs will be rated on their efficacy and a concept will be selected.

6.4.1. Rating system

To make sure an optimal design for the design requirements is selected, some considerations have to be accounted for. Based on the categories named in section 5.4 concepts can be judged on their efficacy. Efficacy will be scored with a rating system. The rating system can be viewed in table 13.

Definition	Rating
System does not meet requirement or is the worst performing concept.	-1
System meets requirement.	0
System exceeds requirement positively or is the best performing concept.	1

Table 13: Rating system for design concept efficacy.

The most applicable system will be the system with the highest score. There are however factors which if not met will make the concept mute. These factors are:

- Thrust
- Safe
- Lightweight

Thrust, however, will be seen as a minimum requirement, meaning that all concepts will be presented in a configuration in which they provide adequate thrust.

6.4.2. Design concepts

Now that the rating system and thresholds are defined. Concepts can be proposed. Three concepts are to be considered.

Concept 1: Powerplant Replacement (PR)

This concept is the easiest proposed system. Since the original platform was capable of all flight in al required directions, the re-motored system should be capable of the same. However, a single change will need to be made to the original design. Since the selected motors will be capable of providing of 0.7 N at 54% thrust a second motor will need to be attached for meeting vertical thrust requirements. However, this system will suffer from constraints in the modifiability of the concept since excess space is not available in quantities.

Concept 2: Frame-Mounted Thrust Vectoring Pods (FMTVP)
The second concept will be a clean slate design which will maximize excess space for modification integration and maintainability. With a box design, a largely empty compartment can be attached to the envelope. The box will also be detachable from the coupling mechanism, which couples the propulsion/ control unit to the envelop. Thus, maximizing ease of maintenance.
Though the original concept is a proven design, it is inefficient for requiring a third motor which is only focused on movement in vertical directions. Therefore, a concept with two motors which can rotate is deemed to be more efficient in regard to weight and electrical consumption.

To allow for the rotation of the motors a vectoring system will need to be designed. This would likely negate weight savings. But two other considerations need to be accounted for, which should contextualize the potential weight penalty. First of all, to move at 1.5 m/s in the vertical direction two specific motors are required, making the weight penalty of the vectoring system mute or potentially a net positive.
In figure 22 a rough concept of the new propulsion system and box design can be viewed.

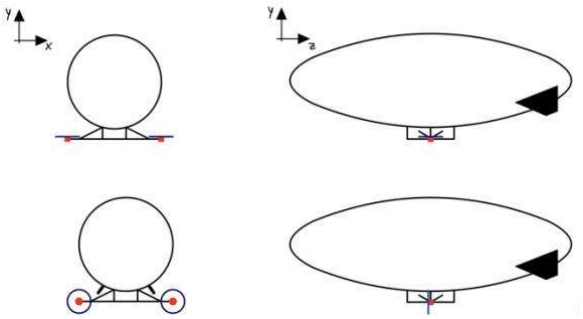


Figure 22: Concept two envisioned while attached to the envelope.

Concept 3: Envelope-Mounted Thrust Vectoring Pods (EMTVP)
The third concept uses the thrust vectoring mount from concept two, while moving the motor mounts to the sides of the envelope.

This setup lengthens the arm between the centre of pressure and the motors. With the longer arm the magnitude of the moment created by the motors will be greater. This, enhances rotational velocity and thus, slightly improves manoeuvrability. The concept makes use of the same box design attached underneath the envelope.

Considering that placement along the sides of the envelope will require wiring to go from the battery pack to the motors, it is considered to be prudent to minimize the distance between the battery and motors. In figure 23 the concept is visualised.

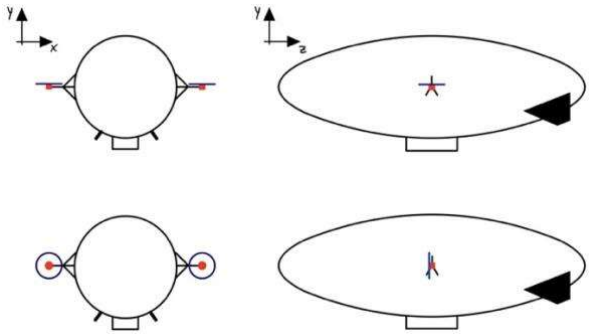


Figure 23: Concept three envisioned while attached to the envelope.

Concept 4: Fixed Motor System (FMS)
Another direction for the propulsion system design is to maximize weigh savings by fixing the motors in one direction. This would mean that the motors would allow movement in one axis. Directional control would be exerted with deflective surfaces on the stabilising fins at the back of the LTAV. This concept can be viewed in figure 24 with the motors mounted at the end of the envelope. While an alteration with the motors mounted in the same position as concept two is displayed in figure 25,

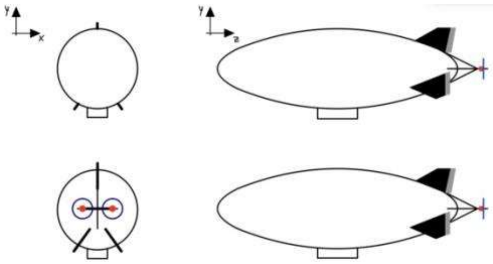


Figure 24: Concept four envisioned while attached to the back end of the envelope.

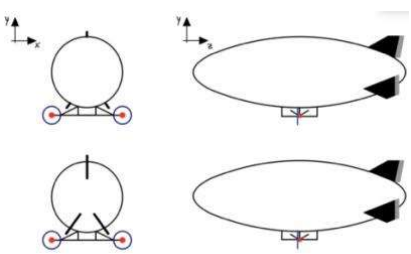


Figure 25: Concept four envisioned while attached bellow the envelope.

This concept will still require three servos unlike other designs which require two or less, adding a bit of weight. Potential weight gains are largely negated by larger control surfaces, similar constructions attached to the envelope and the need for an extra servo and its complimentary wiring.

6.5 Concept selection.
With four concepts laid out in sub-section 6.4.2. and a rating system determined in sub-section 6.4.1. Concepts can be assessed in table 14. The total scores can be viewed in table 15. Based on the total score of 6 points FMTVP is the most advantageous design concept.

Concept	Lightweight	Omni-directional	Multipurpose	Silent	Power Conserving	Maintainable	Safe	Producible
PR	This might be the lightest option since the old frame weighed about 400 grams. Even though two extra motors are added in comparison to the other three concepts, no servos are required. The new motors with propellers mounted will still weigh a third of what the original motors weighed. (1)	Can achieve omni-directionality. However, design requires two additional motors to achieve this omni-directionality. (0)	Frame is aged and has no space for modifications. (-1)	Ads two motors, as noise sources. (-1)	Has four power drawing motors. (-1)	Difficult to work with, electronics are stacked and hard to reach. (-1)	Frame is proven to be safe. However, modifications may result in issues. (0)	Frame is built of two frame sections which are glued together. (1)
FMTVP	Second best option, since everything remains attached to the frame. (0)	Best option due to the thrust vectoring pods. (1)	Box attachment allows for modifications. (1)	Only consists of 2 motors and 2 servos. (1)	Has 2 motors and 2 servos close to the power source minimizing power loss through wiring. (1)	Everything is easy to reach and easily detachable from the airship. (1)	Propeller guards should be considered. (0)	Frame consists of a series of standard-length spanners and rods. (1)
EMTVP	Design ads two independent pods which will	Best option due to the thrust	Box attachment	Only consists of 2	Has 2 motors and 2 servos	Everything is easy to reach. However,	Propeller guards	Frame consists of a series of standard-length spanners

	require their own attachments and lengthier wiring relative to FMTVP and PR. (0)	vectoring pods. (1)	allows for modifications. (1)	motors and 2 servos. (1)	further away from the power source, resulting in more power draw relative to FMTVP. (0)	maintenance to the vectoring pods will require separate detachment form envelope. (0)	should be considered. (0)	and rods. However, the pods and their mounts to the envelope may be more difficult to produce. (0)
FMS	Worst option since three servos are required with large control surfaces. (-1)	Is not omni-directional. (-1)	Box attachment allows for modifications. (1)	Only consists of 2 motors and 3 servos. (0)	Has 2 motors close to the power source but has three servos far from the power source. (0)	Everything is easy to reach and easily detachable from the airship. (1)	Propeller guards should be considered. (0)	Frame consists of a series of standard-length spanners and rods. But servo placement may be complex. (-1)

Table 14: Rationalisation of efficacy scores per design concept.

Concept	Lightweight	Omnidirectional	Multipurpose	Silent	Power Conserving	Maintainable	Safe	Producible	Total
PR	1	0	-1	-1	-1	-1	0	1	**-2**
FMTVP	0	1	1	1	1	1	0	1	**6**
EMTVP	0	1	1	1	0	0	0	0	**3**
FMS	-1	-1	1	0	0	1	0	-1	**-1**

Table 15: Total scores per design concept.

7. Detailing and Realisation

The final two phases, detailing (7.1) and realisation (7,2), actually entail the detail design and operationalisation of the final design.

7.1 Detailing

With the main concept clear detailing of the design will have to commence. First, all already selected components were carefully measured and weighed. After this was concluded CAD, specifically Autodesk Fusion 360, was used to replicate the components in a virtual space. An example of which can be viewed in figure 26, where drawings of the Yinyan BLDC motor can be viewed.

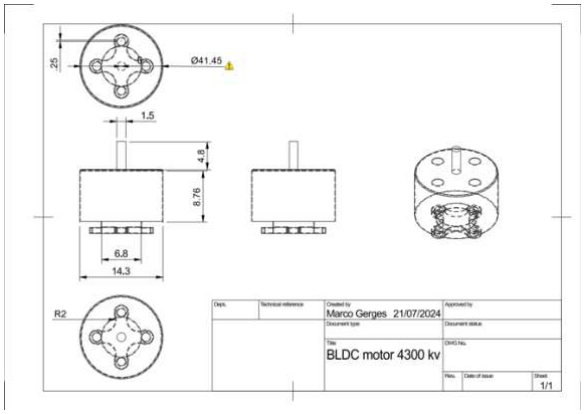


Figure 26: Technical drawing of Yinyan Micro 1104Hm 4300 K_V BLDC motor.

7.1.1. BLDC motor vectoring housing

After concluding the transition of selected components to CAD, the next phase commences with the design of the housing which will be 3D printed for the BLDC motor. This housing is judged to be crucial for determining measurements of further components of the design. The housing can be viewed in figures 27 and 28.

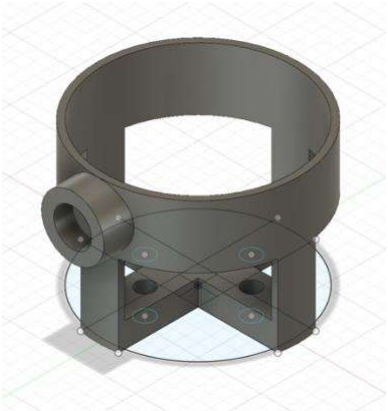


Figure 28: BLDC motor vectoring housing.

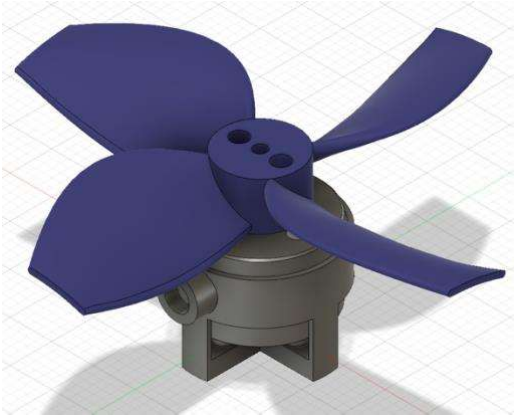


Figure 27: Assembly of motor housing with propeller and BLDC motor.

7.1.2. Electronics and insertions

With a need to rotate this contraption a decision has to be made on what servo to use to rotate this assembly.

Servo

The servo which was selected for this task due to its light weight yet adequate torque is the Power HD HD-1370A, weighing 3.7 g, with a dimension in mm of 20.3x8.5x17.6 and the torque in kilogram centimetres (kg cm) is 0.4. The most important measurement however is the diameter of the shaft which is 3.9 mm. This shaft will be inserted into a tight-fitting carbon fibre tube, run through a bearing while exiting the frame and inserted into the BLDC motor vectoring housing through the extrusion seen in figures 24 and 25. This means that the tube will need to have an internal diameter of 3.9 mm. A quick search online gives a 5mm diameter tube with an internal diameter of 3 mm as the preferred option, since the internal diameter can be expanded to the preferred diameter.

Bearings

Given these measurements a bearing can be selected, this baring needs an internal diameter of 5 mm. A suitable option was found, named the MR105-2RS, weighing in at 1 gram. The bearing also has a thickness of 4 mm giving a good maximum thickness for spanners which will form the frame. This addition, however, will be treated as an optional addition. It is judged that due to the small surface areas and low velocities friction will be negligible.

Batteries

Continuing to the batteries, a flaw in the prior design was the flimsy method of securing the battery to the frame with a strap. This method of securing the battery is risky and though no incidents occurred, the risk is considerable. Conversely, a more modular method of securing battery packs would allow for batteries with different dimensions to be installed and flown with. Considering that all tests occurred with 2S batteries and Voltage is equal when in a parallel circuit, a 2S battery is the optimal option. Considering the original system could fly a maximum of 20 minutes with a 650 milli Ampère hour (mAh) 4S battery weighing 80.8 g, a heavier higher capacity battery is a preferable method of increasing loiter time.

Multiple batteries can be considered in table 16:

Battery	Capacity [mAh]	Weight [g]	Dimensions [mm x mm x mm]
Red Power XT	2200	116	101 x 33.10 x 14.7
Red Power XT	2700	137	101,7 x 33.40 x 27.8
Red Power XT	3500	197	130 x 43.55 x 13,6

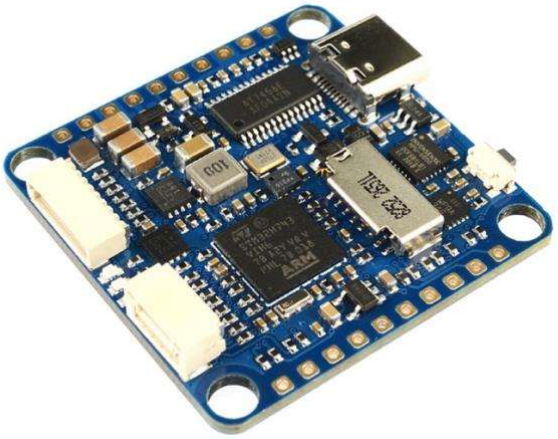
Table 16: 2S batteries with specifications and dimensions.

A battery container which could fit the largest of these batteries with room to spare would allow for modularity and ease of battery replacement. An upside to a container is space for payloads and ballast with easy access. This container will also give space for the Raspberry Pi which is part of the camera setup of Luke Oversluizen. A rough concept of the new propulsion system can be viewed in figure 22.

ESC

A Speed Controller is limited by the minimum amount of current required by the BLDC motor it is supposed to control. There is no limit to how much current capacity is too much for the setup. Another limiting factor is the amount of battery cells. Therefore, an ESC which is compatible with 2S batteries is preferable. Conversely, the AIKON AK32PIN 35A 2-6S 4-IN-1 ESC V2, which was already installed on the original system, can be used.

Flight controller
Similar to ESCs the flight controller is limited by the amount of battery cells. Since the MatekSYS H743-SLIM is very capable of working on a 2S battery the system will be utilised. This will also simplify the control system since INAV is already discussed in this report. Specifications of the Flight controller can be viewed in figure 29 (*Flight Controller H743-SLIM V3 – Matek Systems*, n.d.).



MATEKSYS Flight Controller H743-SLIM

* STM32H743VIH6, 480 MHz, 2MB Flash
* Board V1.0, MPU6000(SPI1) & ICM20602(SPI4)
* Board V1.5, MPU6000(SPI1) & ICM42605(SPI4)
* OSD(SPI2)
* DPS310 (I2C2)
* MicroSD BlackBox (SDIO)
* 7x UARTs, 2x I2C, 1x CAN, 1x SPI3 breakout
* 6x ADC (Vbat, Current, VB2, CU2, RSSI, AirSpeed)
* 13x PWM outputs

* Switchable dual camera inputs
* Switchable 5V/Vbt outputs

* 6~36V DC IN (2~8S LiPo)
* BEC 5V 2A cont. 3A Max.
* Vbat filtered output 1A for VTx and Camera
* LDO 3.3V 200mA cont.

*** ArduPilot MATEKH743
*** BetaFlight MATEKH743
*** INAV MATEKH743

Figure 29: MatekSYS H743-SLIM specifications as provided by MatekSYS on their website.

- Montage: 30,5 x 30,5 mm, Φ4 mm with diameter of 3 mm
- Measurements: 36 x 36 x 5 mm
- Gewicht: 7 g

Camera array
Luke Oversluizen should provide the camera array. However, technical difficulties encountered during the machine learning aspect of the camera project, prevents this report of discussing the camera selection further. It is therefore, still assumed that the camera system will require 200 g. What is known is that a raspberry pi will need to be installed on the HvAirship, therefore, space will be designated for the placement of a raspberry pi.

7.1.3. Frame
Now that all dimensions of components are known the dimensions of the frame can be calculated. First some known measurements will need to be listed. In table 17 all components with weight and measurements can be found.

Component	Dimensions [mm]	Weight [g]
Motor vectoring pod	16	±12
MatekSYS H743-SLIM	36 x 36 x 5	7
Propeller diameter	51	1
AIKON AK32PIN 35A 2-6S 4-IN-1 ESC V2	20 x 20 x 5	10
Largest battery	130 x 43.55 x 13,6	197
Power HD HD-1370A	20.3 x 8.5 x 17.6	3.7
Raspberry Pi 5	85 x 56 x 17	44

Table 17: Dimensions and weights of major sub-components.

Viewing the rough concepts for the proposed propulsion system in figure 30, some interesting considerations can be extrapolated.

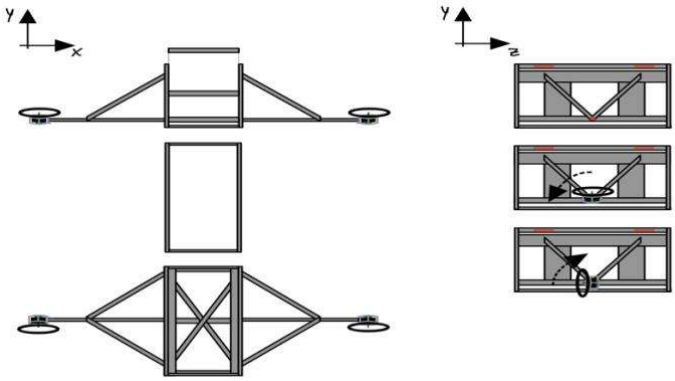


Figure 30: A rough concept of the new propulsion system.

1. There needs to be a margin between the motor vectoring pods and the structural supports. This is to prevent a collision between the propellers and the supports.
2. Servos need to be mounted on the bottom of the frame to minimize their impact on storage capacity.
3. The hight of the frame is dependent on the envelope attachment mount.
4. To maximize maintainability and storage space, electronics will have to be mounted on the sides and outside of the container.

With these considerations measurements of the frame can be extrapolated. With a length of 85 mm the Raspberry Pi is the second longest electronic component after the battery. Since electronics except the battery will need to be installed vertically on the sides of the frame this means that one side of the frame will need to seemingly be at least 90 mm long. However, looking at the Raspberry Pi 5 datasheet dimensions of M2.7 screw holes show that the holes will require 49 x 59. Thus, the maximum side bar Raspberry Pi surface width can be 51.7 x 61.7, rounding this up to 55 x 65 allows for some margin. Based on these dimensions a side plate with ample margins around electronics would be 80 x 65.

It is assessed that the size of the frame itself is of minor importance therefore, a pod which is double the length of the largest battery with some margin. This will give a length of 300 mm. The drawings of the final design can be viewed in figure 31. The drawing is also available in appendix T.

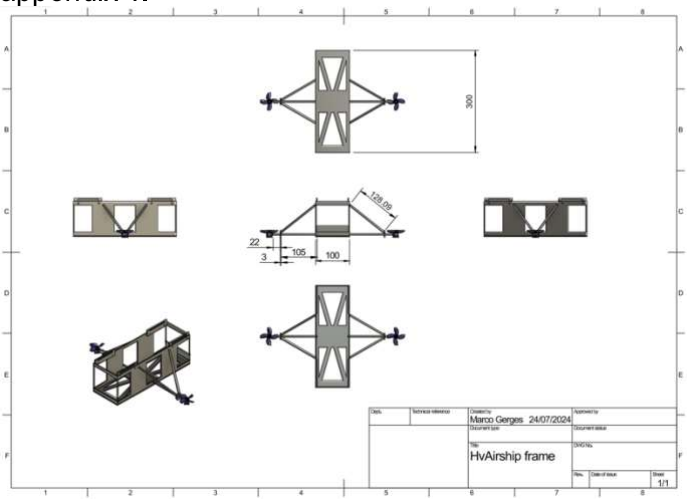


Figure 31: Drawing of the final design.

7.2 Realisation

With the design completed. Some technical aspects need to be discussed. In particular, with regards to the process of filling the envelope with, installing the frame with the propulsion system and the operation of the HvAirship.

7.2.1. Operationalisation guidelines for the HvAirship

The first step when operationalising the HvAirship is to check the envelope for imperfections. When visually inspecting the envelope physically move your fingers over the connection welds between gores. There should be a rough feel to the welds. An example of faulty welds can be viewed in figure 32. Note that the upper two welds are rough and visible while the lower two are barely visible. The lower two in this case are faulty welds.



Figure 32: Example of faulty welds between gores of the envelope.

If this is the case do not use the envelope. Contact Windreiter over the faulty welds. An alternative is to contact the Amsterdam Fashion Institute. The institute has possession of a welder which can weld the envelope temporarily. However, make sure the welder is at the highest welding temperature.

The single most important step in the operationalisation of the HvAirship is filling the envelope correctly. Making sure the envelope is not damaged during the filling process is of great importance. It is advised to utilise a polyvinyl chloride (PVC) pipe as an intermediate between the pressure tank nipple and the envelope. This prevents the cold temperature of the He gas from damaging the envelope. When filling the envelope, flow must be restricted as much as possible.

When the envelope is filled, it is advised to attach the frame attachment while connected to the frame. This to ensure the frame is able to be connected, which may not be ensured when connecting the attachments alone. Use aviation tape through the four holes in the envelope attachments and roll the tape all around the envelope.

Detach the frame from the envelope attachments and attach weights to clarify the actual lifting capacity. When lifting capacity is extrapolated, consult ballast weight chart in appendix U to make the HvAirship slightly negatively buoyant. This slightness can be viewed as the HvAirship floating in the air while still moving to any flow of air surrounding it. The HvAirship should also slowly move down towards the ground.

Connect the batteries to the flight controller and listen for a beeping sound from the motors. When this sound stops, turn on the remote control and check if INAV has established a connection. If this is not the case trigger the left switch until the display states that the HvAirship is connected.

Make sure no objects on board are unsecured.

7.2.2. Operating the HvAirship

When operating the HvAirship make sure to avoid sharp surfaces. If the balloon moves too quickly check if the INAV settings are set correctly. If not, lower the power output to 54% and check if the HvAirship still moves to fast. If this is the case lower the power output in 10% increments.

Please note that the HvAirship only has permission to fly if it minimizes noise output. Therefore, it is crucial for the power output to be set as low as possible. Conversely, avoid flying within 3 meters of students and staff. Maintain a distance of 1 meter minimum to walls, windows and the roof.

Use the fourth floor E wing area which extends out in front of the fifth floor as the emergency landing area. The HvAirship is programmed to fly towards this area when detecting issues.

It is of utmost importance that the envelope remains filled with He. If creases start to increase on the envelope immediately refill the envelope as discussed in sub-section 7.2.1.

If the HvAirship is to be grounded longer than one week, during which time the envelope is not accessible for refill the envelope should immediately be emptied of He, using a vacuum. Use one of the spare envelopes as a guide on how to fold the envelope properly.

Make sure at least two reserve envelopes are available for the first operational HvAirship and per additional operational airship add one envelope to this number.

When modifying the HvAirship is not possible due to size constraints of the frame. Use frame section DXF files to laser cut or STL files to 3D print additional sections for expansions. The standardized plate formats should allow the frame to be semi-modular.

Do not introduce hydrogen into the envelope, though lifting capacity may increase slightly, the HvAirship is not allowed to fly with hydrogen inside JMH as per correspondence discussed in section 4.3.

Discussion

Over the course of this project, shortcomings in methodology and limitations in the capabilities of the AUAS' technical capacities and the student's capability were uncovered. These shortcomings and limitations have limited the academic proficiency of the thesis to a degree and therefore need to be discussed.

Time

A major constraint from the start of the project, and one that had been discussed during the planning phases as well, was time. Due to the late start of the project and the need of capabilities only accessible in the JMH, which is closed for a portion of the year, time was a constant source of pressure.

Massive difficulties were encountered due to periods where time was wasted while waiting, examples of which are:

- A six-week period lost due to technical issues with envelopes.
- Two weeks spent waiting for technical data to be compiled.
- Days spent waiting for components and/or sensors to arrive.

These delays sadly, resulted in the fact that the student was not able to build a prototype before the writing of this report and the summer recess.

Sensory Reliability

A major shortcoming in the methodology of the project was the use of a single sensor for sound intensity readings. The sound intensity values for the ambient sound in JMH and the sound intensity measures of the propulsion systems, are conversely not validated with a redundant sensory system. The readings were also manually performed using a sound intensity meter instead of analog/ digital recordings, which would have been more reliable.

Communication

Communication from the student towards the lecturer and lab lead was sparse and often negative due to technical difficulties encountered. The stubborn attitude displayed by the student when encountering issues and slow attempts at seeking help lost valuable time for the project. A major learning point is that communication will always have to take precedent over attempts at achieving progress.

Scientific unknowns

A large unknown during this project was the effect of the helium balloon on the noise and vibrations transmitted by the propulsion system. Logic dictates that the envelope would act as an amplifier. However, due to the early grounding of the HvAirship due to the initial leak and extended grounding caused by technical faults in replacement envelopes combined with the lack of sound intensity meters while the HvAirship was still capable of flight, prevented any effort of an analysis being performed on this interesting aspect. The Sorama camera was offered by the lab lead at the start of the project was not utilised, to the regret of the student.

Positives

It must be credited however that a design has been proposed. This after multiple technical setbacks, difficulties with literature scarcity and time constraints, shows the large amount of effort put in to achieve the objectives. It has to be noted that the small amount of lifting capacity afforded by the envelope dimensions, which themselves were limited by the dimensions of the atrium of JMH, was not the only technical issue. Noise intensity had to be tackled as well as aerodynamic questions which due to the lack of prevalence of airships in science for the last ninety years had to be inferred from modifications on particle physics formulas for ellipsoid shapes.

Conclusion

This design report was created to modify/ redesign the HvAirship. A lighter than air remote controlled vehicle which is supposed to allow students to perform experiments and achieve learning objectives by modifying, utilising and/ or flying the platform for IP-projects.

The main research question was designated as:

'What changes can be made to the propulsion system of the HvAirship drone for it to be quieter while at least maintaining, if not improving its manoeuvrability?'

This main research query was answered in five stages with five sub-questions. These stages and sub-questions were based on the five stages of design projects. Namely, orientation, analysis, design, detailing and realisation. The sub-questions are:

1. What stakeholders in the HvAirship project can be identified?
2. What are the design requirements for the propulsion system of the HvAirship?
3. What are relevant propulsion methods for the HvAirship while accounting for design requirements?
4. What is the most applicable propulsion method for the current predicament?
5. How will the designed propulsion system be integrated onto the HvAirship and its functions?

The stakeholders were analysed and as a result the maintenance lab lead was designated as the most powerful direct stakeholder. For the long term for the HvAirship project the Facility Services of the Jakoba Mulderhuis (JMH) are identified as the most likely future obstacle to continuation of the project and should be treated as such.

The design requirements were identified by conducting interviews with the main stakeholder and can be summarised as the maximalisation of loiter time and a velocity of 3 meters per second (m/s) in the longitudinal axis and 1.5 m/s vertically. The HvAirship should also be able to manoeuvre in the aforementioned directions from a stationary position. The propulsion system itself should allow for future modifications and minimize its impact on the carrying capacity of the envelope.

To identify the noise impact of the HvAirship, tests were conducted to find the sound intensity and frequency of the original HvAirship configuration. Sound intensity in the JMH without the HvAirship were also measured from which a minimum mean noise intensity of 55 decibels was calculated for quieter days. This value was selected as the maximum sound intensity considered to be acceptable for the future propulsion system of the HvAirship.

Relevant propulsion systems were assessed and eventually a conventional propulsion system in tandem with a vectoring mount was selected as the most applicable design direction. The Yinyan 1104hm brushless DC (BLDC) motor in tandem with the GemFan 51-millimetre diameter four bladed propellers were proven to be capable of complying with the aforementioned design requirements.

The final design is a dual motor pod design connected to a modular frame which is attached below the envelope.

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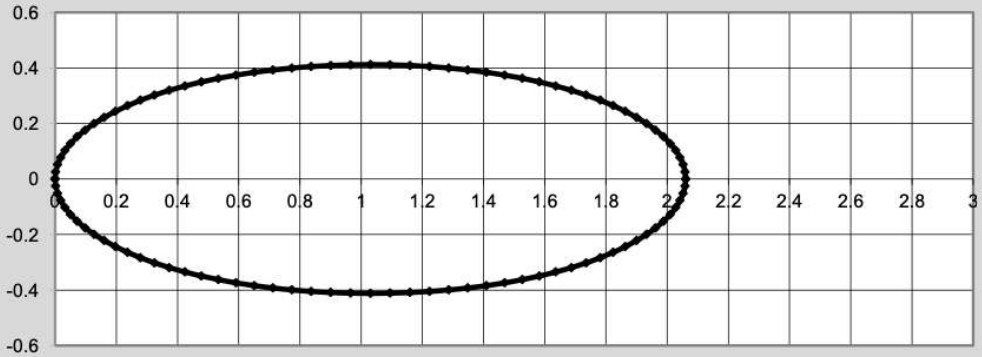
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Appendix A: Windreiter 4-Gore Elliptical Balloon specifications sheet.

Airship Envelope Datasheet			
Identifier:	4 - Gore Elliptical Balloon	Date of Design:	2023-04-19
Designed by:		Date of Production:	
Organisation:		Place of Production:	Windreiter HQ



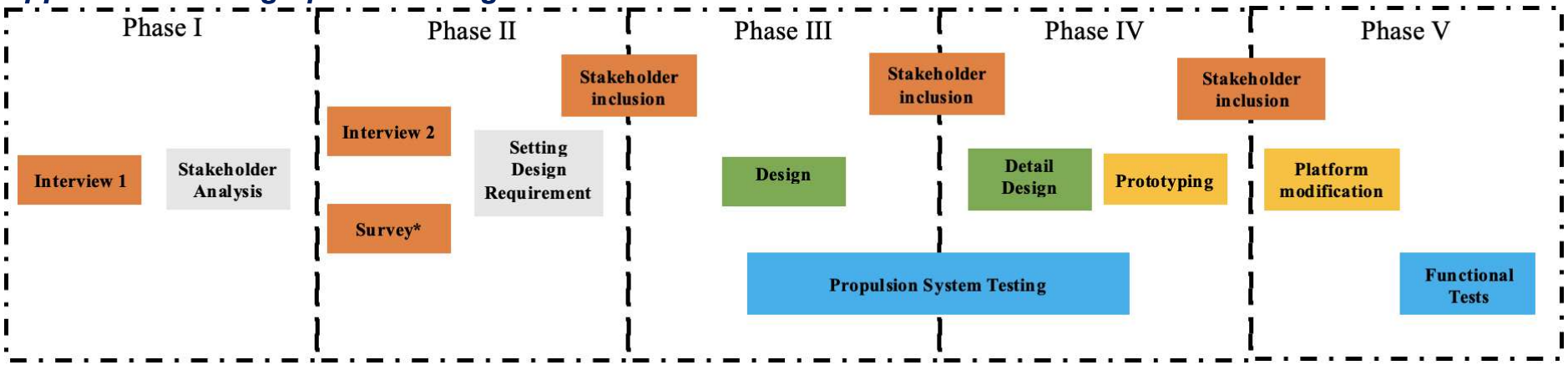
Material Parameter		Gertler Shape Coefficients	
Envelope Material:	TritaX Silver	Position of max. thickness:	0.5
Surface Weight [g/m²]:	30	Bow Radius:	0.5
Tension Strength [MPa]:	245	Stern Radius:	0.5
Bonding Technique:	Point Welding	Prismatic Coefficient:	0.67

Design Parameter		Lift Assumptions*	
Length to Diameter:	2.500	Lift at Sea Level He [g]:	600.40
Block Volume [m³]:	1.400	Lift at 300 m Helium [g]:	578.41
Block Coefficient:	0.526	Lift at Sea Level H2 [g]:	647.01
Envelope Volume [l]:	736.70	Lift at 300 m H2 [g]:	623.31
Length [m]:	2.06	*The calculations assume pure lifting gas	
Diameter [m]:	0.82		
Surface Area [m²]:	4.54		
Envelope Weight [g]:	136.30		
Number of gores	4		

Gertler 4621 Shape spreadsheet. Computes shape, volume, centre of volume, and pattern for a Series58 Model 4621 body of revolution.Copyright (C) 2019 Johannes Eissing

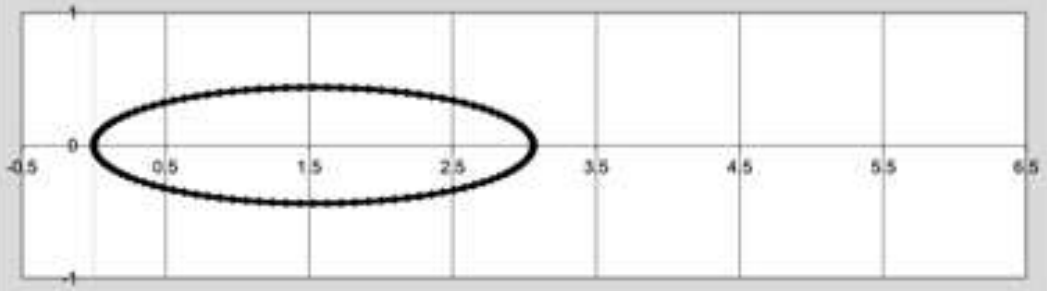
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Single envelopes to large numbers
www.windreiter.de

Appendix B: Design process diagram



Appendix C: Datasheet provided by Windreiter HQ for new envelope.

Airship Envelope Datasheet			
Identifier:	3m Elliptical	Date of Design:	2024-06-25
Designed by:		Date of Production:	
Organisation:	HVA	Place of Production:	Windreiter HQ



Material Parameter		Gertler Shape Coefficients	
Envelope Material:	TritaX Silver	Position of max. thickness:	0.5
Surface Weight [g/m²]:	30	Bow Radius:	0.5
Tension Strength [MPa]:	245	Stern Radius:	0.5
Bonding Technique:	Point Welding	Prismatic Coefficient:	0.667

Design Parameter		Lift Assumptions*	
Length to Diameter:	3.500	Lift at Sea Level He [g]:	1024
Block Volume [m³]:	2.350	Lift at 300 m Helium [g]:	986
Block Coefficient:	0.524	Lift at Sea Level H2 [g]:	1103
Envelope Volume [l]:	1231	Lift at 300 m H2 [g]:	1063
Length [m]:	3.06	*The calculations assume pure lifting gas	
Diameter [m]:	0.88		
Surface Area [m²]:	6.92		
Envelope Weight [g]:	208		
Number of gores	4		

Gertler 4621 Shape spreadsheet. Computes shape, volume, centre of volume, and pattern for a Series58 Model 4621 body of revolution. Copyright (C) 2019 Johannes Eissing

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Appendix D: Tables with drag coefficient values per velocity in the longitudinal and vertical axes.

Assumed thrust correction margin	20%

Horizontal								
U [m/s]	v [km/h]	Re	C_D	T [N]	T [g]	T_c [g]	T_c2 [N]	T_c2 [g]
1	3,6	2,019E+05	0,38	0,1	14,29	17,2	0,08	8,6
1,1	3,96	2,221E+05	0,38	0,2	17,25	20,7	0,10	10,4
1,2	4,32	2,423E+05	0,38	0,2	20,49	24,6	0,12	12,3
1,3	4,68	2,624E+05	0,38	0,2	23,99	28,8	0,14	14,4
1,4	5,04	2,826E+05	0,38	0,3	27,77	33,3	0,16	16,7
1,5	5,4	3,028E+05	0,38	0,3	31,82	38,2	0,19	19,1
1,6	5,76	3,230E+05	0,38	0,4	36,14	43,4	0,21	21,7
1,7	6,12	3,432E+05	0,38	0,4	40,73	48,9	0,24	24,4
1,8	6,48	3,634E+05	0,38	0,4	45,58	54,7	0,27	27,4
1,9	6,84	3,836E+05	0,38	0,5	50,71	60,9	0,30	30,4
2	7,2	4,038E+05	0,38	0,6	56,11	67,3	0,33	33,7
2,1	7,56	4,239E+05	0,38	0,6	61,77	74,1	0,36	37,1
2,2	7,92	4,441E+05	0,37	0,7	67,70	81,2	0,40	40,6
2,3	8,28	4,643E+05	0,37	0,7	73,90	88,7	0,44	44,3
2,4	8,64	4,845E+05	0,37	0,8	80,37	96,4	0,47	48,2
2,5	9	5,047E+05	0,37	0,9	87,11	104,5	0,51	52,3
2,6	9,36	5,249E+05	0,37	0,9	94,11	112,9	0,55	56,5
2,7	9,72	5,451E+05	0,37	1,0	101,38	121,7	0,60	60,8
2,8	10,08	5,653E+05	0,37	1,1	108,91	130,7	0,64	65,3
2,9	10,44	5,854E+05	0,37	1,1	116,71	140,1	0,69	70,0
3	10,8	6,056E+05	0,37	1,2	124,78	149,7	0,73	74,9
3,1	11,16	6,258E+05	0,37	1,3	133,11	159,7	0,78	79,9
3,2	11,52	6,460E+05	0,37	1,4	141,71	170,1	0,83	85,0
3,3	11,88	6,662E+05	0,37	1,5	150,58	180,7	0,89	90,3
3,4	12,24	6,864E+05	0,37	1,6	159,71	191,6	0,94	95,8
3,5	12,6	7,066E+05	0,37	1,7	169,10	202,9	1,00	101,5
3,6	12,96	7,268E+05	0,37	1,8	178,76	214,5	1,05	107,3
3,7	13,32	7,469E+05	0,37	1,9	188,69	226,4	1,11	113,2
3,8	13,68	7,671E+05	0,37	2,0	198,88	238,7	1,17	119,3
3,9	14,04	7,873E+05	0,37	2,1	209,33	251,2	1,23	125,6
4	14,4	8,075E+05	0,37	2,2	220,05	264,1	1,30	132,0

Vertical								
U [m/s]	v [km/h]	Re	C_D	T [N]	T [g]	T_c [g]	T_c2 [N]	T_c2 [g]
1	3,6	5,806E+04	0,39	0,5	50,23	60,3	0,30	30,1
1,1	3,96	6,386E+04	0,39	0,6	60,86	73,0	0,36	36,5
1,2	4,32	6,967E+04	0,39	0,7	72,48	87,0	0,43	43,5
1,3	4,68	7,547E+04	0,39	0,8	85,10	102,1	0,50	51,1
1,4	5,04	8,128E+04	0,39	1,0	98,71	118,5	0,58	59,2
1,5	5,4	8,708E+04	0,39	1,1	113,30	136,0	0,67	68,0
1,6	5,76	9,289E+04	0,39	1,3	128,87	154,6	0,76	77,3
1,7	6,12	9,870E+04	0,39	1,4	145,43	174,5	0,86	87,3
1,8	6,48	1,045E+05	0,39	1,6	162,96	195,5	0,96	97,8
1,9	6,84	1,103E+05	0,39	1,8	181,46	217,8	1,07	108,9
2	7,2	1,161E+05	0,39	2,0	200,94	241,1	1,18	120,6
2,1	7,56	1,219E+05	0,39	2,2	221,39	265,7	1,30	132,8
2,2	7,92	1,277E+05	0,39	2,4	242,80	291,4	1,43	145,7
2,3	8,28	1,335E+05	0,39	2,6	265,19	318,2	1,56	159,1
2,4	8,64	1,393E+05	0,39	2,8	288,54	346,2	1,70	173,1
2,5	9	1,451E+05	0,39	3,1	312,85	375,4	1,84	187,7
2,6	9,36	1,509E+05	0,39	3,3	338,12	405,7	1,99	202,9
2,7	9,72	1,568E+05	0,39	3,6	364,36	437,2	2,14	218,6
2,8	10,08	1,626E+05	0,38	3,8	391,56	469,9	2,30	234,9
2,9	10,44	1,684E+05	0,38	4,1	419,71	503,7	2,47	251,8
3	10,8	1,742E+05	0,38	4,4	448,82	538,6	2,64	269,3

Appendix E: Ambient noise intensity measurements with basic statistical data.

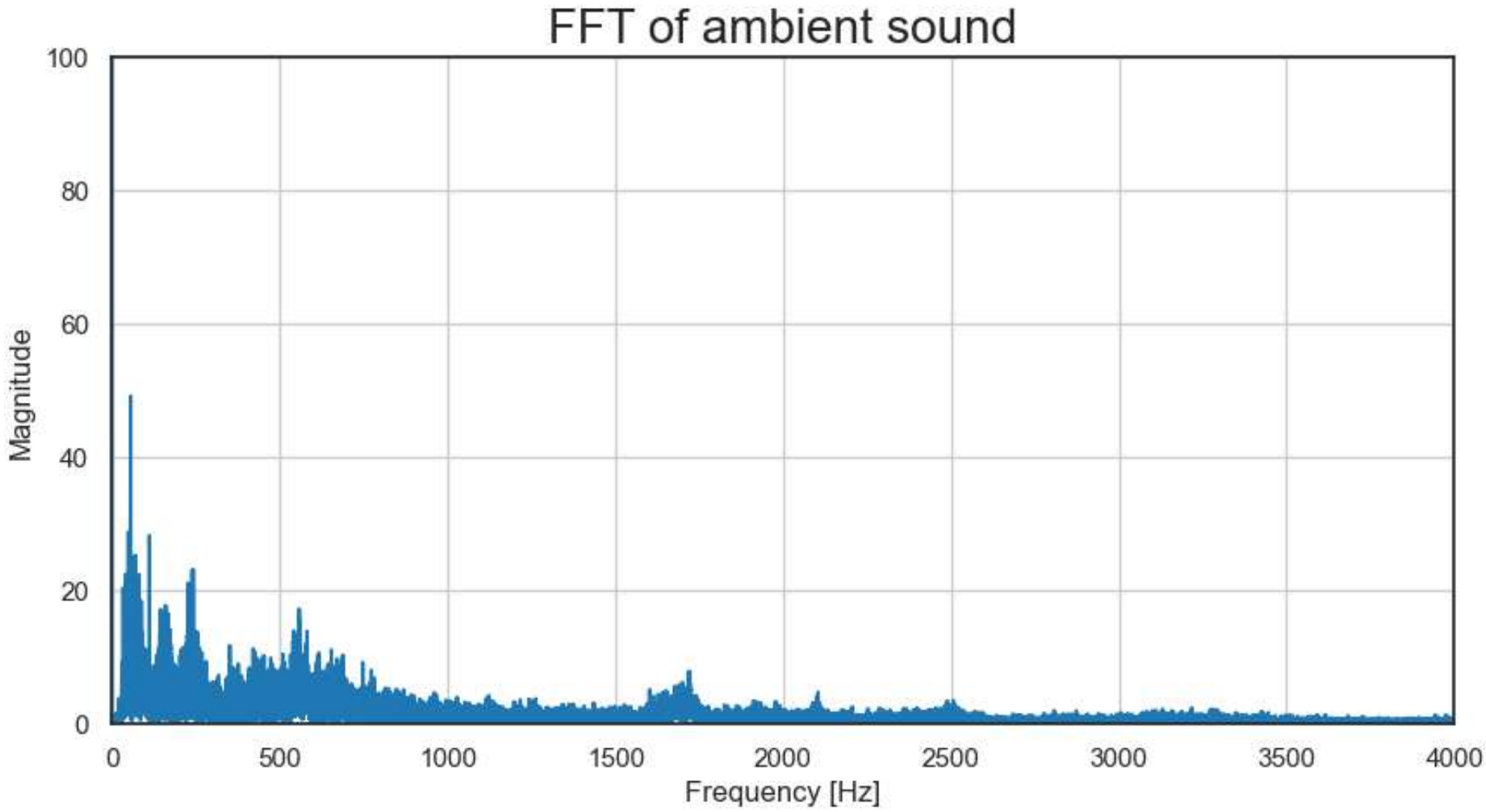
Tue	04-06-2024										
Location/time	08:00	09:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00	17:00	18:00
1	53	65	57	63	70	63	73	66	62	58	49
2	53	63	61	64	66	67	74	65	59	54	50
3	57	64	62	62	63	65	70	72	63	61	52
4	51	65	58	60	67	68	68	69	61	68	51
5	52	61	63	62	65	66	67	67	60	57	50
Comment											

Fri	07-06-2024										
Location/time	08:00	09:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00	17:00	18:00
1	49	52	55	61	60	59	56	53	49	47	
2	47	53	62	63	62	61	57	56	51	50	
3	51	50	60	63	61	60	60	55	52	51	
4	48	57	58	57	59	59	54	50	48	49	
5	47	56	57	58	58	57	55	57	47	48	
Comment											Building was closing down

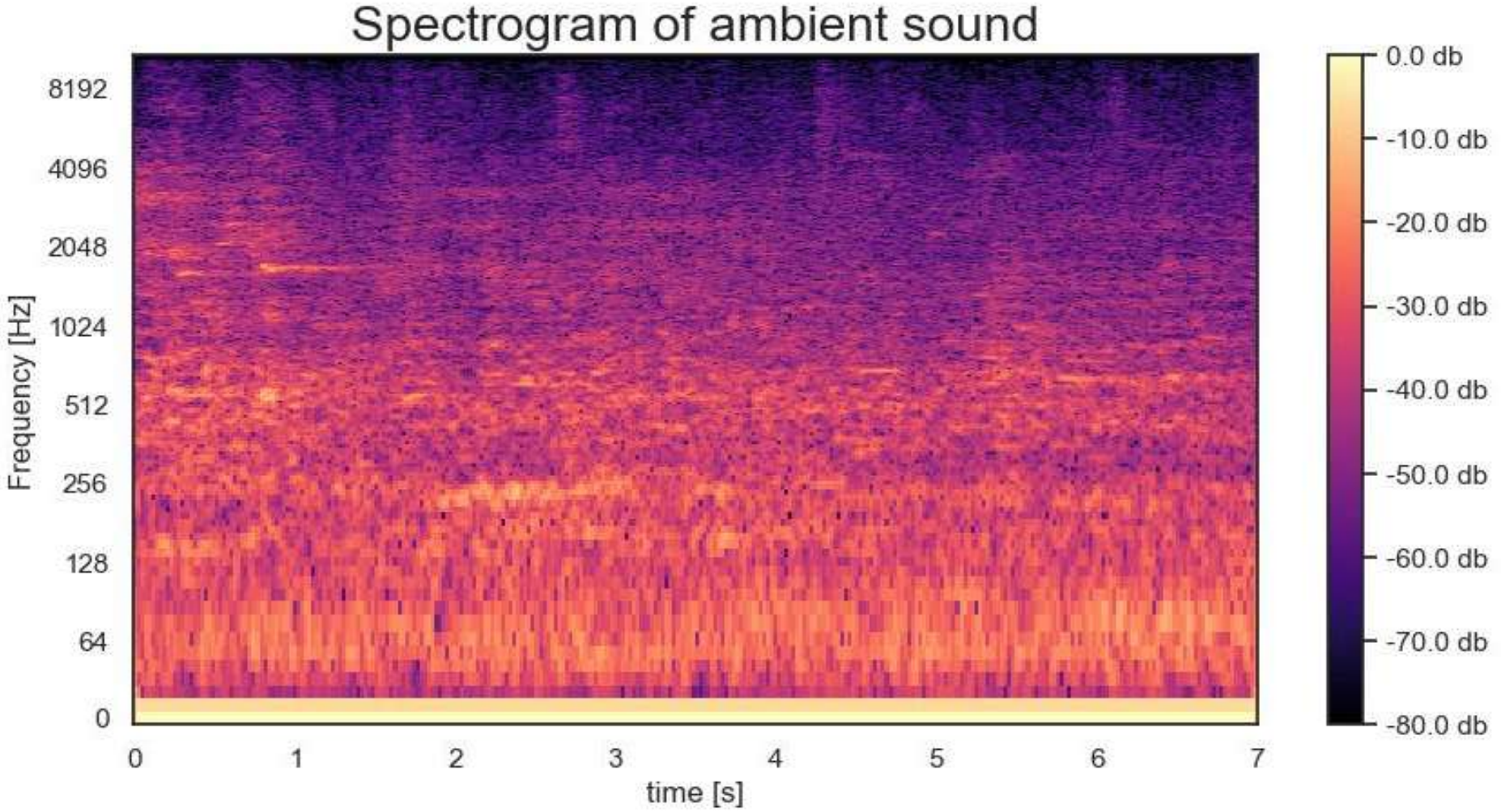
Tuesday Statistics			
Average	Mean	Maximum	Minimum
62	63	74	49

Friday Statistics			
Mean	Median	Maximum	Minimum
55	56,5	63	47

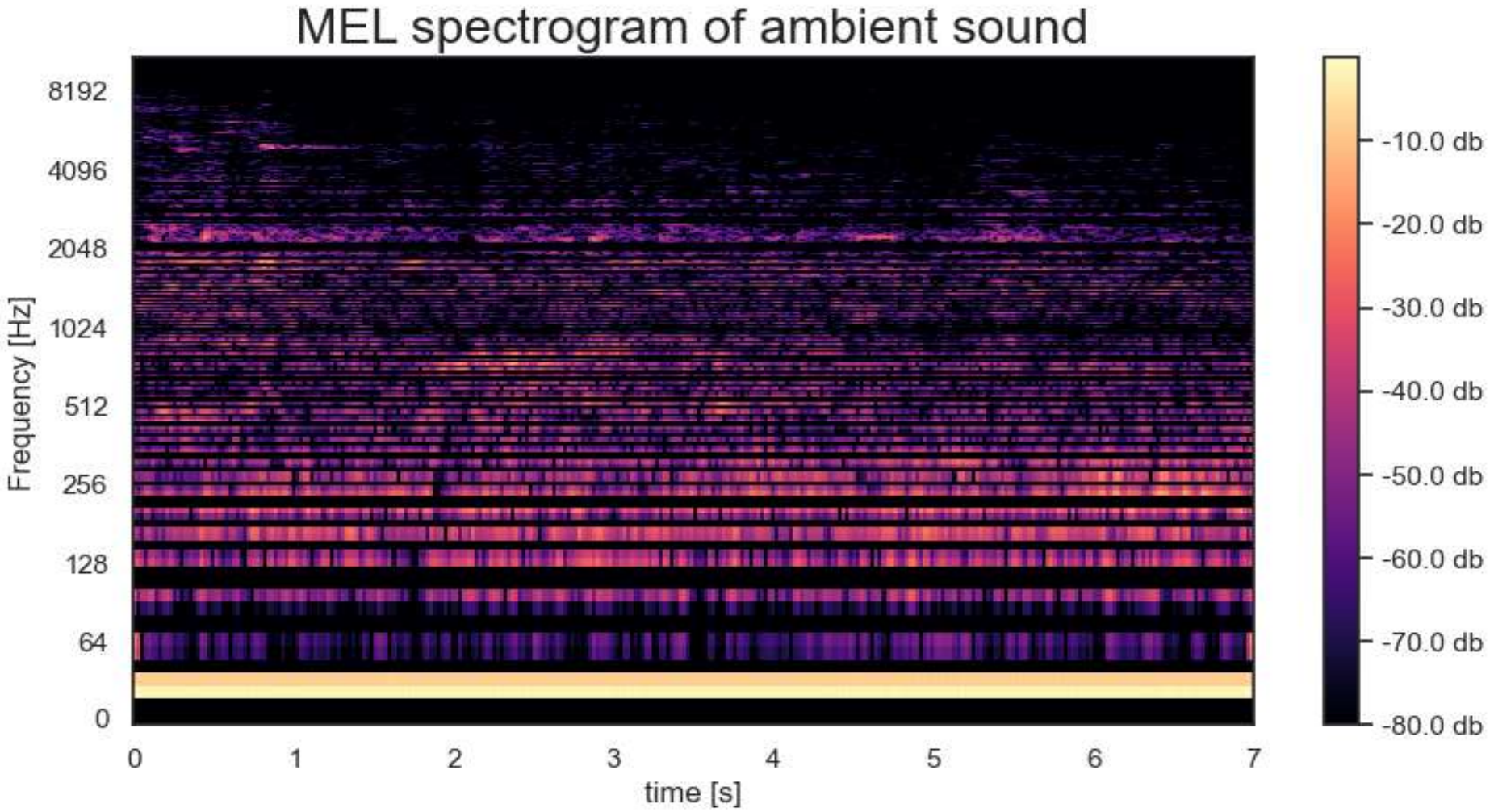
Appendix F: FFT of ambient sound.



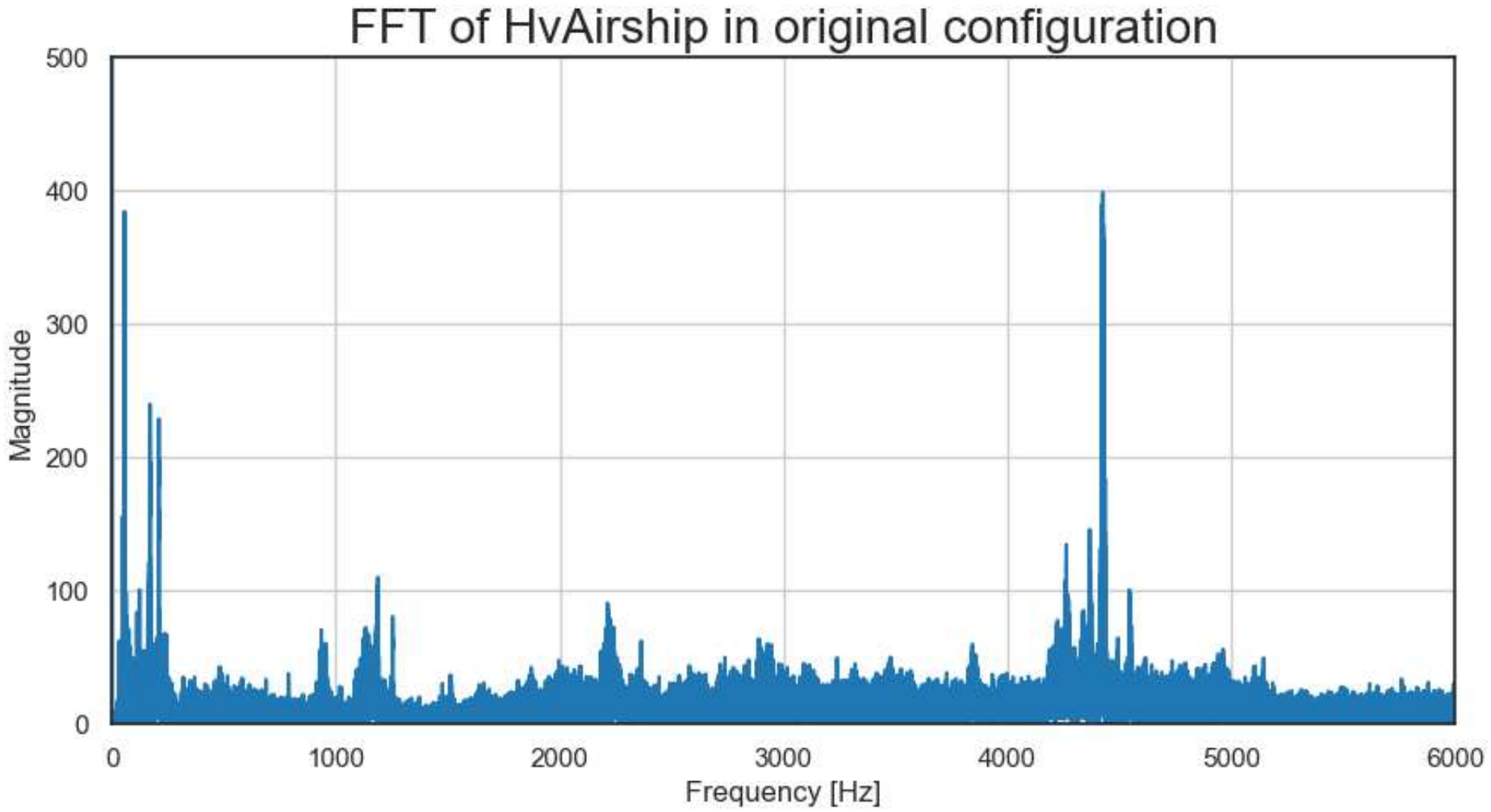
Appendix G: Spectrogram of ambient sound



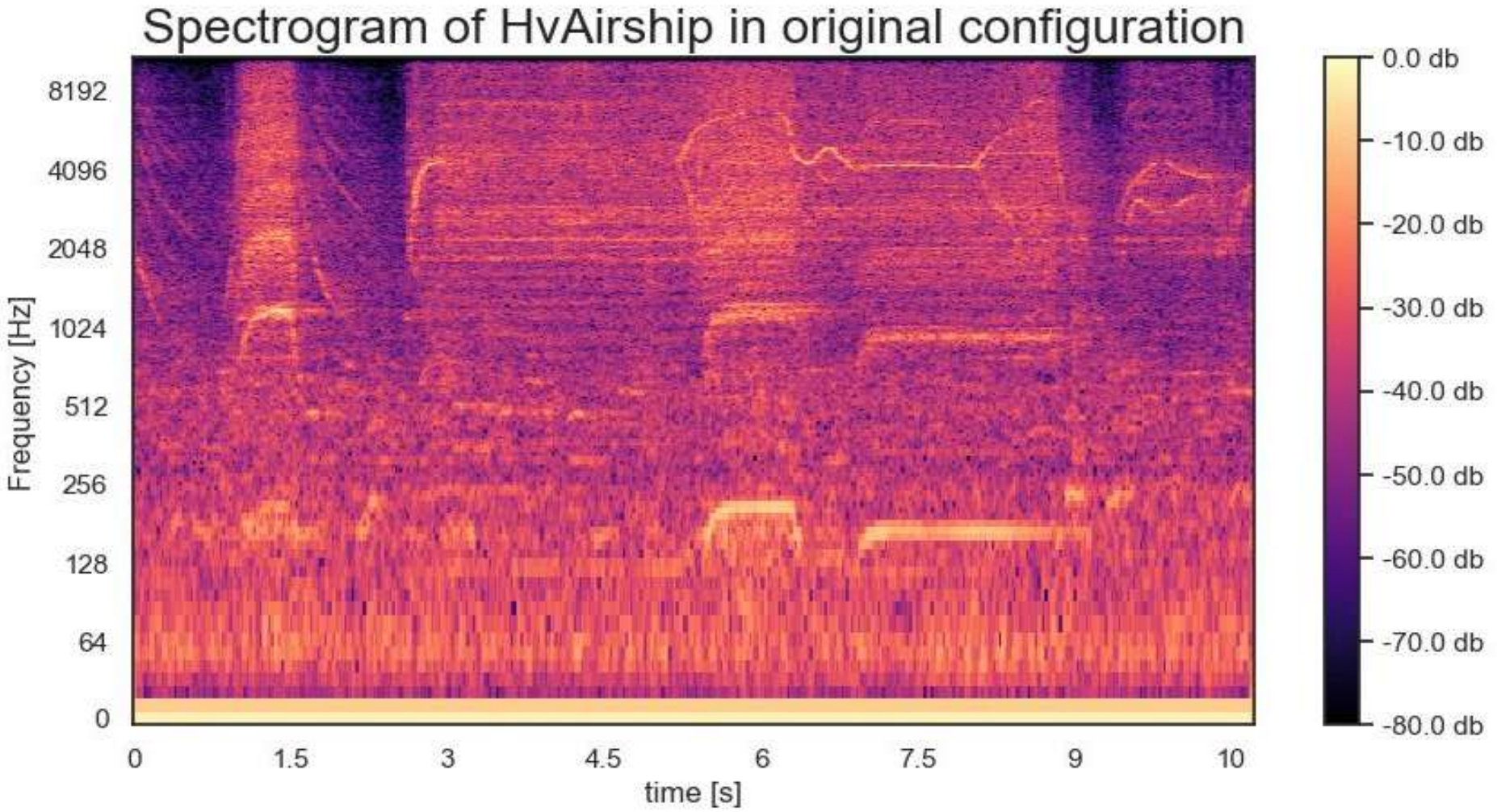
Appendix H: MEL Spectrogram of ambient sound.



Appendix I: FFT of HvAirship in original configuration.



Appendix J: Spectrogram of HvAirship in original configuration.



Appendix K: MEL Spectrogram of HvAirship in original configuration.

